

MAT211: Calculus-II

Lecture Notes

Instructor: Emon Hossain

April 16, 2026

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1 Lecture-01

Question: what did we mean by the derivative $f'(a)$?

Answer: Let U be an open subset of $\mathbb{R}$ and $f : U \rightarrow \mathbb{R}$ a function. Then f is differentiable at a , with derivative m , if and only if

$$\lim_{h \rightarrow 0} \frac{1}{h} \left(\underbrace{f(a+h) - f(a)}_{\Delta f} - \underbrace{mh}_{(*)} \right) = 0 \quad (1)$$

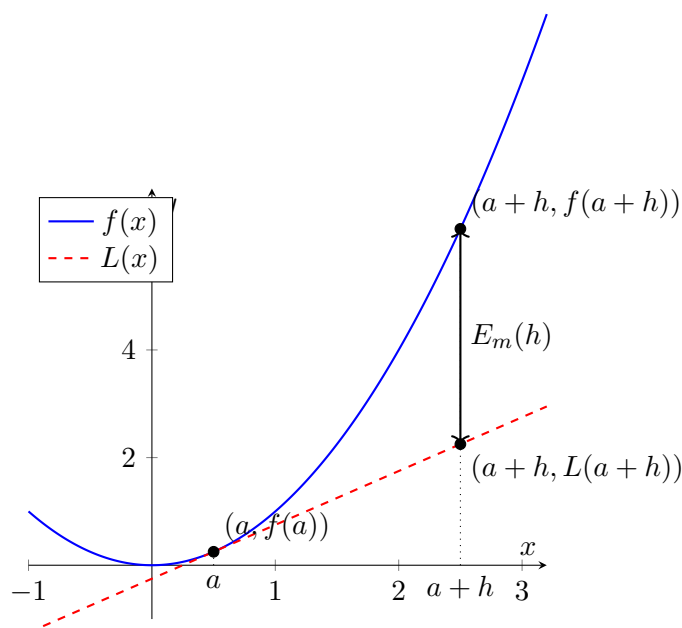
$(*)$: linear function of $\Delta x = h$.

Question: Why impose the domain U that needs to be an open set?

Answer: Write the answer.

The function mh that multiplies h by the derivative m is thus a linear function of the change in x . This definition emphasizes the idea that a function f is differentiable at a point a if the increment Δf to the function is well approximated by a linear function of the increment h to the variable, this linear function is $f'(a)h$. Let's see how we can confirm that:

Suppose $f(x)$ is our desired function and we want to approximate the function in the neighborhood at $x = a$. For this purpose, we will draw several straight lines through the point $(a, f(a))$ and see which line approximates the function best. Here, the best means, the linear line should approximate the function well at $x = a + h$. Suppose the slope of that linear function is m .



Then,

$$\begin{aligned} \frac{L(a+h) - L(a)}{a+h-a} &= m \\ \frac{L(a+h) - f(a)}{h} &= m \\ L(a+h) &= f(a) + mh \end{aligned}$$

Now, we want to minimize the error, $E_m(h) = f(a+h) - L(a+h)$. Hence, we need to find the optimal value for m , for which:

$$\frac{E_m(h)}{h} \rightarrow 0$$

$$\boxed{\text{optimal } m \iff \lim_{h \rightarrow 0} \frac{E_m(h)}{h} = 0}$$

A simple computation show that,

$$\lim_{h \rightarrow 0} \frac{E_m(h)}{h} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - mh}{h} = 0$$

which is the desired condition 1, we showed in the definition.

2 Lecture-02

2.1 Vector Spaces

Definition 2.1 (Euclidean Vector Space). A Euclidean vector space is a finite-dimensional inner product space over $\mathbb{R}$.

Definition 2.2 (Inner product space). An inner product space is a vector space V over the field F together with an inner product, that is, a map

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow F$$

that satisfies the following three properties for all vectors $x, y, z \in V$ and all scalars $a, b \in F$.

1. (Symmetry) $\langle x, y \rangle = \langle y, x \rangle$.
2. (Linearity in the first argument) $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$.
3. (Positive-definiteness) If $x \neq 0$ then, $\langle x, x \rangle > 0$.

Example 2.3. Verify that the dot product function $\langle \cdot, \cdot \rangle : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is an inner product.

To verify this we must prove the three conditions shown above. Let $u = (a, b)$ and $v = (c, d)$. Then,

1. $\langle u, v \rangle = ac + bd = ca + db = \langle v, u \rangle$.
2. Let $a, b \in \mathbb{R}$ and $u, v, w \in \mathbb{R}^2$ with $u = (u_1, u_2), v = (v_1, v_2), w = (w_1, w_2)$. Then

$$\begin{aligned} \langle au + bv, w \rangle &= (au_1 + bv_1)w_1 + (au_2 + bv_2)w_2 \\ &= au_1w_1 + bv_1w_1 + au_2w_2 + bv_2w_2 \\ &= a(u_1w_1 + u_2w_2) + b(v_1w_1 + v_2w_2) \\ &= a\langle u, w \rangle + b\langle v, w \rangle. \end{aligned}$$

3. For any $u = (u_1, u_2)$,

$$\langle u, u \rangle = u_1^2 + u_2^2.$$

Since squares of non-zero numbers are positive, we have $u_1^2 > 0, u_2^2 > 0$ and therefore, $\langle u, u \rangle > 0$.

Definition 2.4 (Informal). Define **norm** of a vector $x \in V$ as

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

2.2 Metric Spaces

Definition 2.5 (Metric Space). A metric space is an ordered pair (M, d) where M is a set and d is a **metric** (distance) on M , i.e., a function

$$d : M \times M \rightarrow \mathbb{R}$$

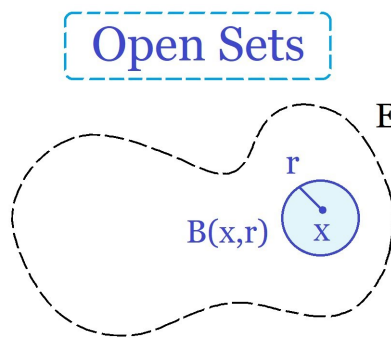
satisfying the following axioms for all points $x, y, z \in M$:

1. $d(x, y) \geq 0$;
2. $d(x, y) = 0 \iff x = y$;
3. $d(x, y) = d(y, x)$.
4. $d(x, z) \leq d(x, y) + d(y, z)$.

Example 2.6 (Taxi-cab Metric). In $\mathbb{R}^2$, the taxi-cab distance between $p = (p_1, p_2)$ and $q = (q_1, q_2)$ is $|q_1 - p_1| + |q_2 - p_2|$. It is easy to verify that this is a metric.

Definition 2.7 (Open ball). For any $x \in \mathbb{R}^n$ and any $r > 0$, the **open ball** of radius r around x is the subset

$$B_r(x) = \{y \in \mathbb{R}^n \mid |x - y| < r\}.$$



Definition 2.8 (Open sets of $\mathbb{R}^n$). A subset $U \subseteq \mathbb{R}^n$ is open if for every point $x \in U$, there exists $r > 0$ such that the open ball $B_r(x)$ is contained in U .

Example 2.9. We prove that the Cartesian product $(a_1, b_1) \times (a_2, b_2)$ of two open intervals in $\mathbb{R}$ is an open set in $\mathbb{R}^2$ with the standard Euclidean topology.

Proof. Let d denote the Euclidean metric on $\mathbb{R}^2$:

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}.$$

Take an arbitrary point $p = (x, y) \in (a_1, b_1) \times (a_2, b_2)$. Then

$$a_1 < x < b_1, \quad a_2 < y < b_2.$$

Define the four positive distances to the endpoints:

$$d_1 = x - a_1, \quad d_2 = b_1 - x, \quad d_3 = y - a_2, \quad d_4 = b_2 - y.$$

Let

$$r = \min\{d_1, d_2, d_3, d_4\} > 0.$$

We claim that the open ball $B(p, r) = \{q \in \mathbb{R}^2 : d(p, q) < r\}$ is contained in $(a_1, b_1) \times (a_2, b_2)$.

Indeed, take any point $q = (u, v) \in B(p, r)$. Then

$$|u - x| \leq d(p, q) < r, \quad |v - y| \leq d(p, q) < r.$$

Hence

$$x - r < u < x + r, \quad y - r < v < y + r.$$

Because $r \leq x - a_1$ and $r \leq b_1 - x$, we have

$$a_1 \leq x - r < u < x + r \leq b_1,$$

so $u \in (a_1, b_1)$. Similarly, $r \leq y - a_2$ and $r \leq b_2 - y$ imply

$$a_2 \leq y - r < v < y + r \leq b_2,$$

so $v \in (a_2, b_2)$. Consequently $q = (u, v) \in (a_1, b_1) \times (a_2, b_2)$.

Thus every point of the product has an open neighbourhood (an open ball) contained in the product. By definition, $(a_1, b_1) \times (a_2, b_2)$ is open in $\mathbb{R}^2$. $\square$

Definition 2.10 (Closed set). A closed set A is defined by its ability to contain all of its limit points, meaning if a sequence of points $\{x_n\}$ within A converges to a limit x , then that limit x must also be in A .

Definition 2.11 (Sequence). A sequence is a function whose domain is $\mathbb{N}$.

3 Lecture-03

3.1 Sequences and Convergence

Definition 3.1 (Sequence). A sequence is a function where domain is $\mathbb{N}$.

Example 3.2. For example, let $x_n = \frac{1}{n}$ or $\{\frac{1}{n}\}_{n=1}^{\infty}$, be a sequence. Here, we write the sequence $f : \mathbb{N} \rightarrow \mathbb{R}$, where $f(n) = x_n = \frac{1}{n}$.

Definition 3.3 (Convergence of a Sequence). A sequence of real numbers converges to a real number a if for every number $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$

$$|a_n - a| < \varepsilon$$

Example 3.4. Take our earlier example 3.2, prove that $x_n = \{\frac{1}{n}\}_{n=1}^{\infty}$ converges to 0.

Proof. Let, $\varepsilon > 0$, we choose $N \in \mathbb{N}$ such that $N > 1/\varepsilon$ (which is possible due to the Archimedean property), giving us $\frac{1}{N} < \varepsilon$.

Then, for all $n \geq N$ we have

$$\left| \frac{1}{n} - 0 \right| \leq \left| \frac{1}{N} - 0 \right| < \varepsilon$$

□

Example 3.5. Consider $x_{n+1} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right)$, $x_0 > 0$

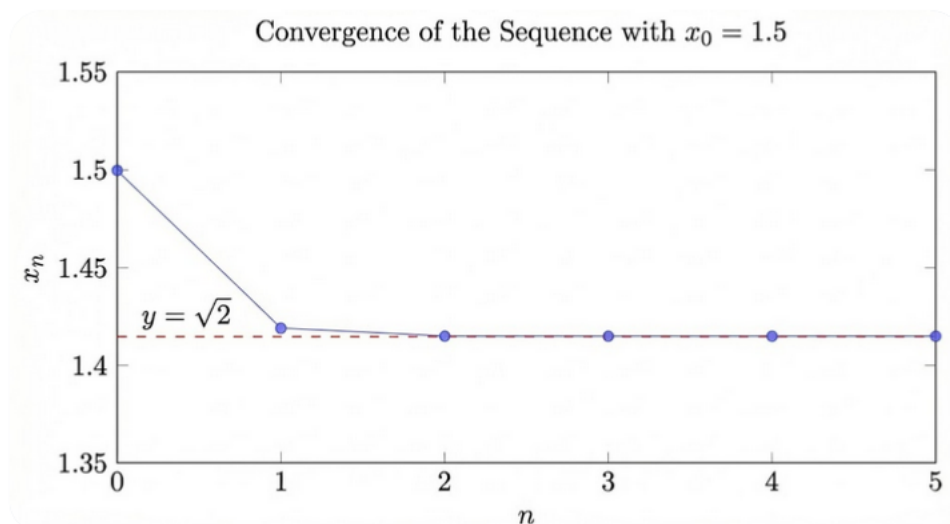
Show that x_n converges.

Proof. First, note that $x_n > 0$ for all $n \in \mathbb{N}$. Because, $x_0 > 0$ and for induction, if we assume that $x_n > 0$ then,

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right) > 0$$

since all the terms in left side is positive.

Then note that, if you start above $\sqrt{2}$ you stay above and decrease. Refer to the picture below for an example.



Next, to rigorously proof that the limit exists. We can do so by proving this sequence to be monotone and bounded. We can prove that by testing both cases $0 < x_0 \leq \sqrt{2}$ and $0 < x_0 < \sqrt{2}$. Now, let $(x_n) \rightarrow L$, where $L > 0$. Then $(x_{n+1}) \rightarrow L$

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} x_{n+1} \\ &= \lim_{n \rightarrow \infty} \frac{1}{2} \left(x_n + \frac{2}{x_n} \right) \\ &= \lim_{n \rightarrow \infty} \frac{x_n}{2} + \frac{1}{x_n} \\ &= \frac{L}{2} + \frac{1}{L} \end{aligned}$$

This gives us,

$$L = \frac{L}{2} + \frac{1}{L} \implies \frac{L}{2} = \frac{1}{L} \implies L = \sqrt{2}$$

Hence, the given sequence converges to $\sqrt{2}$. □

Definition 3.6 (Convergent sequence; limit of sequence for $\mathbb{R}^n$ Space). A sequence $i \rightarrow a_i$ of points $\mathbb{R}^n$ converges to $a \in \mathbb{R}^n$ if

$$\forall \varepsilon > 0, \exists M | m > M \implies |a_m - a| < \varepsilon$$

We then call a , the *limit* of the sequence

This also holds for vector a_i .

$$a_m = [(a_m)_1, (a_m)_2, \dots, (a_m)_n] \rightarrow [a_1, a_2, \dots, a_n] \in \mathbb{R}^n$$

Here, a_1 is the convergent point of sequence $(a_m)_1, (a_{m+1})_1, \dots$. And so on.

Proposition 3.7 (Convergence in terms of coordinates). A sequence $m \rightarrow a_m$ with $a_m \in \mathbb{R}^n$ converges to a if and only if each coordinate; i.e. for all $1 \leq j \leq n$, the j th coordinate $(a_i)_j$ of a_i converges to a_j , the j th coordinate of the limit a .

Proof. Let $\|\cdot\|$ denote the standard Euclidean norm in $\mathbb{R}^n$. Recall the definition of sequence convergence: $a_m \rightarrow a$ means that for every $\epsilon > 0$, there exists an integer N such that for all $m \geq N$, $\|a_m - a\| < \epsilon$.

Forward Direction ($\Rightarrow$):

Assume $a_m \rightarrow a$. We want to show that for any fixed coordinate $j \in \{1, \dots, n\}$, the sequence $(a_m)_j \rightarrow a_j$.

Let $\epsilon > 0$ be given. Because $a_m \rightarrow a$, there exists an integer N such that for all $m \geq N$:

$$\|a_m - a\| < \epsilon$$

Observe that the absolute value of any single coordinate difference is bounded by the Euclidean distance:

$$|(a_m)_j - a_j| = \sqrt{((a_m)_j - a_j)^2} \leq \sqrt{\sum_{k=1}^n ((a_m)_k - a_k)^2} = \|a_m - a\|$$

Therefore, for all $m \geq N$, we have:

$$|(a_m)_j - a_j| \leq \|a_m - a\| < \epsilon$$

Since for any $\epsilon > 0$ we found an N that satisfies this condition, $(a_m)_j \rightarrow a_j$ for each j .

Reverse Direction ($\Leftarrow$):

Assume that for each $1 \leq j \leq n$, the coordinate sequence $(a_m)_j \rightarrow a_j$. We want to show that the vector sequence $a_m \rightarrow a$.

Let $\epsilon > 0$ be given. By the definition of convergence for each coordinate, there exists an integer N_j such that for all $m \geq N_j$:

$$|(a_m)_j - a_j| < \frac{\epsilon}{\sqrt{n}}$$

We want this to hold for *all* coordinates at the same time, so we choose the largest threshold: let $N = \max\{N_1, N_2, \dots, N_n\}$.

For any $m \geq N$, m is greater than or equal to every N_j , so the inequality $|(a_m)_j - a_j| < \frac{\epsilon}{\sqrt{n}}$ holds for all $j = 1, \dots, n$ simultaneously.

Now, we evaluate the distance between the vectors a_m and a for $m \geq N$:

$$\|a_m - a\| = \sqrt{\sum_{j=1}^n ((a_m)_j - a_j)^2} < \sqrt{\sum_{j=1}^n \left(\frac{\epsilon}{\sqrt{n}}\right)^2}$$

Simplifying the expression under the square root:

$$\|a_m - a\| < \sqrt{\sum_{j=1}^n \frac{\epsilon^2}{n}} = \sqrt{n \cdot \frac{\epsilon^2}{n}} = \sqrt{\epsilon^2} = \epsilon$$

Because for any $\epsilon > 0$ we found an N such that $m \geq N \implies \|a_m - a\| < \epsilon$, the sequence a_m converges to a . $\square$

4 Lecture-04

4.1 Scalar Functions of Several Variables

In many areas of mathematics, we encounter quantities that depend on more than one variable. For instance, the temperature in a room may depend on position, the elevation of a landscape depends on location, and the pressure of a gas may depend on both volume and temperature. These are all examples of *scalar functions of several variables*: functions that assign a real number to each point in $\mathbb{R}^n$.

Definition 4.1. A *scalar function of several variables* is a function

$$f : \mathbb{R}^n \rightarrow \mathbb{R}.$$

Thus, to each point

$$\vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n,$$

the function assigns a real number $f(\vec{x})$.

In practice, instead of writing

$$f\left(\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}\right),$$

it is more convenient to write simply

$$f(x_1, \dots, x_n) \text{ or } f\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

This is only a change in notation; the meaning is the same.

Definition 4.2. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. The *graph* of f is the subset of $\mathbb{R}^{n+1}$ defined by

$$\text{graph}(f) = \left\{ \begin{pmatrix} \vec{x} \\ f(\vec{x}) \end{pmatrix} : \vec{x} \in \mathbb{R}^n \right\}.$$

Equivalently, the graph is the set of all points $(x_1, \dots, x_n, f(x_1, \dots, x_n))$ in $\mathbb{R}^{n+1}$.

When $n = 1$ or $n = 2$, the graph often gives a useful geometric picture of the function. For functions of two variables, one may think of the graph as a surface in $\mathbb{R}^3$.

Another helpful way to visualize a function is through its *level sets*. For a constant $c \in \mathbb{R}$, the level set of f corresponding to c is the set

$$\{\vec{x} \in \mathbb{R}^n : f(\vec{x}) = c\}.$$

In the case $n = 2$, these are often called *level curves* or *contour curves*, since they resemble the contour lines on a topographic map.

Example 4.3. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{4}}$$

Find the domain and range of f , and describe its graph and level curves for various values of $c = 0, 1$.

Example 4.4. Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = x^2 + y^2.$$

Then f assigns to each point (x, y) the square of its distance from the origin.

Its graph is the set

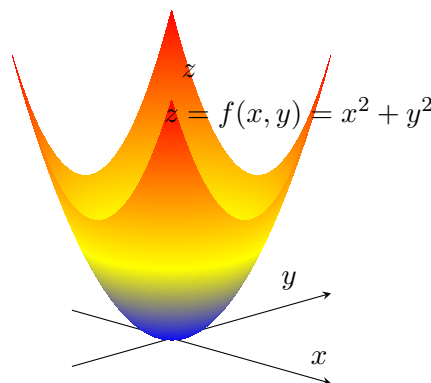
$$\text{graph}(f) = \{(x, y, x^2 + y^2) : (x, y) \in \mathbb{R}^2\},$$

which is a paraboloid in $\mathbb{R}^3$.

For each constant $c \geq 0$, the level curve $f(x, y) = c$ is given by

$$x^2 + y^2 = c,$$

which is a circle centered at the origin with radius $\sqrt{c}$.



4.2 Vector Functions of Several Variables

In many mathematical settings, the quantity we wish to study is not a single real number but a vector. For example, the velocity of a moving particle has several components, a force field assigns a vector to each point in space, and a parametrized curve records position as a vector depending on one or more variables. This leads naturally to the notion of a *vector-valued function of several variables*.

Such functions take points in $\mathbb{R}^n$ as inputs and return vectors in $\mathbb{R}^m$. In practice, they may be viewed either as geometric mappings or as collections of scalar component functions. Both points of view are useful, and we will move freely between them.

Definition 4.5. A *vector function of several variables* is a function

$$\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

Thus, for each point

$$\vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n,$$

the value $\mathbf{f}(\vec{x})$ is a vector in $\mathbb{R}^m$.

If

$$\mathbf{f}(\vec{x}) = \begin{pmatrix} f_1(\vec{x}) \\ \vdots \\ f_m(\vec{x}) \end{pmatrix},$$

then each $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is a scalar-valued function, called the *i*th *component function* of $\mathbf{f}$. In this way, a vector-valued function may be regarded as a collection of m scalar-valued functions of n variables.

It is often convenient to write the input vector in coordinate form. Thus, instead of writing $\mathbf{f}(\vec{x})$, where

$$\vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

we may simply write

$$\mathbf{f}(x_1, \dots, x_n).$$

This is only a notational simplification.

Definition 4.6. Let $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a vector-valued function. If

$$\mathbf{f}(\vec{x}) = \begin{pmatrix} f_1(\vec{x}) \\ \vdots \\ f_m(\vec{x}) \end{pmatrix},$$

then $\mathbf{f}$ is said to be determined by its component functions

$$f_1, \dots, f_m : \mathbb{R}^n \rightarrow \mathbb{R}.$$

Accordingly, many properties of $\mathbf{f}$ can be studied by examining its components one at a time.

Vector-valued functions are especially important in geometry and applied mathematics. They arise in the study of parametrized curves and surfaces, as well as in vector fields such as velocity fields, gravitational fields, and electromagnetic fields.

Example 4.7. Define $\mathbf{f} : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by

$$\mathbf{f}(u, v) = \begin{pmatrix} u \\ v \\ u^2 + v^2 \end{pmatrix}.$$

Then $\mathbf{f}$ is a vector-valued function of two variables.

Its component functions are

$$f_1(u, v) = u, \quad f_2(u, v) = v, \quad f_3(u, v) = u^2 + v^2.$$

Geometrically, $\mathbf{f}$ assigns to each point (u, v) in the plane the point

$$(u, v, u^2 + v^2)$$

in $\mathbb{R}^3$. Thus, $\mathbf{f}$ parametrizes the paraboloid

$$z = x^2 + y^2.$$

Example 4.8. Define $\mathbf{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$\mathbf{F}(x, y) = \begin{pmatrix} -y \\ x \end{pmatrix}.$$

This is a vector field in the plane. At each point (x, y) , the function assigns the vector $(-y, x)$.

Its component functions are

$$F_1(x, y) = -y, \quad F_2(x, y) = x.$$

This field represents a rotational pattern about the origin, since each vector is perpendicular to the position vector (x, y) .

For more interactive visualizations of scalar and vector functions, see https://mathinsight.org/level_sets.

4.3 Limits of Functions of One Variable

Definition 4.9. Let I be an open interval containing c , and let f be a function defined on I , except possibly at c . We say that the *limit* of $f(x)$ as x approaches c is L , and write

$$\lim_{x \rightarrow c} f(x) = L,$$

if for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x \neq c$,

$$|x - c| < \delta \implies |f(x) - L| < \varepsilon.$$

Example 4.10. Show that

$$\lim_{x \rightarrow 4} \sqrt{x} = 2.$$

Solution:

Before we use the formal definition, let us first try some numerical tolerances. Suppose the y -tolerance is 0.5, or $\varepsilon = 0.5$. How close to 4 does x have to be so that y is within 0.5 units of 2, that is, so that $1.5 < y < 2.5$? Since $y = \sqrt{x}$, we proceed as follows:

$$1.5 < y < 2.5$$

$$1.5 < \sqrt{x} < 2.5$$

$$1.5^2 < x < 2.5^2$$

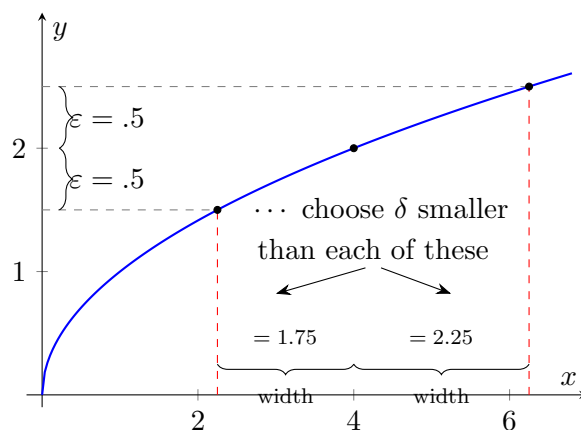
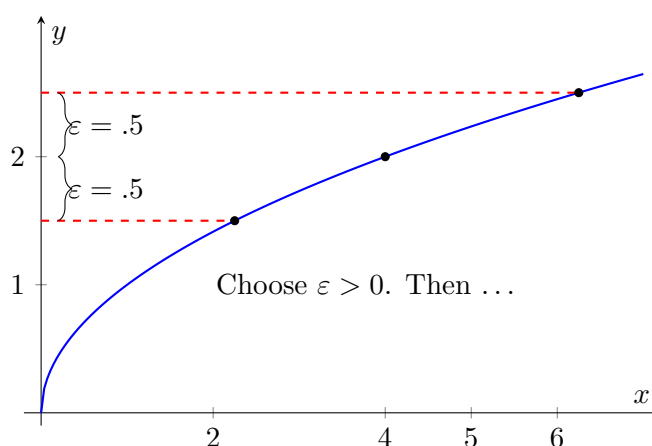
$$2.25 < x < 6.25.$$

So, what is the desired x -tolerance? We want a symmetric interval of x -values, namely

$$4 - \delta < x < 4 + \delta.$$

The lower bound 2.25 is 1.75 units from 4, while the upper bound 6.25 is 2.25 units from 4. We need the smaller of these two distances, so we must have

$$\delta \leq 1.75.$$



With $\varepsilon = 0.5$, we pick any $\delta < 1.75$.

Given the y tolerance $\varepsilon = 0.5$, we have found an x tolerance, $\delta \leq 1.75$, such that whenever x is within δ units of 4, then y is within ε units of 2. That is exactly what we wanted.

Let us try another value of ε .

What if the y tolerance is 0.01, that is, $\varepsilon = 0.01$? How close to 4 does x have to be in order for y to be within 0.01 units of 2 (or $1.99 < y < 2.01$)? Again, we square these values to obtain

$$1.99^2 < x < 2.01^2,$$

or

$$3.9601 < x < 4.0401.$$

What is the desired x tolerance? In this case, we must have

$$\delta \leq 0.0399,$$

which is the smaller of the two distances from 4 to the endpoints above.

Notice that there appear to be two possible tolerances: below 4 we have 0.0399 units, while above 4 we have 0.0401 units. However, we cannot use the larger value 0.0401 for δ , because then the interval for x would be

$$3.9599 < x < 4.0401,$$

which would produce some y -values outside the interval of points within 0.01 units of 2.

So far we have found that if $\varepsilon = 0.5$, then $\delta \leq 1.75$, and if $\varepsilon = 0.01$, then $\delta \leq 0.0399$. Since the pattern is not yet obvious, we now switch to a general ε and determine δ symbolically.

Assume that $y = \sqrt{x}$ is within ε units of 2:

$$\begin{aligned} |y - 2| &< \varepsilon \\ -\varepsilon &< y - 2 < \varepsilon && \text{(definition of absolute value)} \\ -\varepsilon &< \sqrt{x} - 2 < \varepsilon && \text{(since } y = \sqrt{x}\text{)} \\ 2 - \varepsilon &< \sqrt{x} < 2 + \varepsilon && \text{(add 2)} \\ (2 - \varepsilon)^2 &< x < (2 + \varepsilon)^2 && \text{(square all terms)} \\ 4 - 4\varepsilon + \varepsilon^2 &< x < 4 + 4\varepsilon + \varepsilon^2 && \text{(expand)} \\ 4 - (4\varepsilon - \varepsilon^2) &< x < 4 + (4\varepsilon + \varepsilon^2). \end{aligned}$$

This last inequality is in the desired form

$$4 - \text{something} < x < 4 + \text{something}.$$

Since we want this interval to describe an x tolerance centered at 4, we must choose δ to be the smaller of the two quantities:

$$\delta \leq 4\varepsilon - \varepsilon^2 \quad \text{or} \quad \delta \leq 4\varepsilon + \varepsilon^2.$$

Therefore the appropriate choice is

$$\delta \leq 4\varepsilon - \varepsilon^2.$$

Since $\varepsilon > 0$, the minimum is $\delta \leq 4\varepsilon - \varepsilon^2$. Thus the formula is: given an ε , set

$$\delta \leq 4\varepsilon - \varepsilon^2.$$

We can check this against our previous values. If $\varepsilon = 0.5$, then the formula gives

$$\delta \leq 4(0.5) - (0.5)^2 = 1.75,$$

and if $\varepsilon = 0.01$, then the formula gives

$$\delta \leq 4(0.01) - (0.01)^2 = 0.0399.$$

So, given any $\varepsilon > 0$, choose $\delta \leq 4\varepsilon - \varepsilon^2$. Then if $|x - 4| < \delta$ (and $x \neq 4$), we have

$$|f(x) - 2| < \varepsilon,$$

satisfying the definition of the limit. Therefore we have shown formally that

$$\lim_{x \rightarrow 4} \sqrt{x} = 2.$$

Actually, this is a bit inconvenient if $\varepsilon \geq 4$, since then $4\varepsilon - \varepsilon^2 \leq 0$. Although ε is usually thought of as small, this case can still occur. To avoid this issue, one may simply choose

$$\delta = \min\{1, 4\varepsilon - \varepsilon^2\}$$

when $0 < \varepsilon < 4$, and take $\delta = 1$ when $\varepsilon \geq 4$.

Example 4.11. Prove directly from the ε - δ definition that

$$\lim_{x \rightarrow 2} x^2 = 4.$$

Proof. Let $\varepsilon > 0$ be given. We must show that there exists a number $\delta > 0$ such that

$$0 < |x - 2| < \delta \implies |x^2 - 4| < \varepsilon.$$

The key observation is that

$$|x^2 - 4| = |x - 2||x + 2|.$$

Thus the problem is that, besides the small factor $|x - 2|$, there is also the extra factor $|x + 2|$, which depends on x . If we could somehow control $|x + 2|$ by a constant, then we could finish the argument.

At first glance, one might try to require

$$|x - 2| < \frac{\varepsilon}{|x + 2|},$$

but this does not immediately solve the problem, because δ is supposed to be a fixed number and may not depend on x .

To obtain a constant bound, assume first that

$$\delta < 1.$$

Then from

$$|x - 2| < \delta$$

it follows that

$$|x - 2| < 1.$$

Hence

$$-1 < x - 2 < 1,$$

so

$$1 < x < 3.$$

Adding 2 gives

$$3 < x + 2 < 5.$$

Therefore,

$$|x + 2| < 5.$$

Equivalently, since $3 < x + 2 < 5$, taking reciprocals gives

$$\frac{1}{5} < \frac{1}{|x + 2|} < \frac{1}{3},$$

and so

$$\frac{\varepsilon}{5} < \frac{\varepsilon}{|x + 2|}.$$

This shows that a sufficient condition is

$$|x - 2| < \frac{\varepsilon}{5}.$$

Accordingly, let us choose

$$\delta = \min \left\{ 1, \frac{\varepsilon}{5} \right\}.$$

Now suppose that

$$0 < |x - 2| < \delta.$$

Then, since $\delta \leq 1$, we have $|x - 2| < 1$, and thus $|x + 2| < 5$. Also, since $\delta \leq \varepsilon/5$, we have

$$|x - 2| < \frac{\varepsilon}{5}.$$

Therefore,

$$|x^2 - 4| = |x - 2| |x + 2| < \frac{\varepsilon}{5} \cdot 5 = \varepsilon.$$

Thus, whenever $0 < |x - 2| < \delta$, it follows that $|x^2 - 4| < \varepsilon$. Hence

$$\lim_{x \rightarrow 2} x^2 = 4.$$

□

The above examples are taken from [https://math.libretexts.org/Bookshelves/Calculus/Calculus_3e_\(Apex\)/01%3A_Limits/1.02%3A_Epsilon-Delta_Definition_of_a_Limit](https://math.libretexts.org/Bookshelves/Calculus/Calculus_3e_(Apex)/01%3A_Limits/1.02%3A_Epsilon-Delta_Definition_of_a_Limit). For more examples like these, visit the link above.

5 Lecture-05

5.1 Limits of Functions of multiple Variables

Definition 5.1. Let $U \subset \mathbb{R}^n$ be an open subset containing a neighborhood of $a \in \mathbb{R}^n$, except perhaps for the point a itself. Suppose $f : U \rightarrow \mathbb{R}^m$. We say that

$$\lim_{x \rightarrow a} f(x) = \ell$$

($f(x)$ approaches $\ell \in \mathbb{R}^m$ as x approaches a) if for every $\varepsilon > 0$ there is $\delta > 0$ so that

$$\|f(x) - \ell\| < \varepsilon \quad \text{whenever} \quad 0 < \|x - a\| < \delta.$$

5.2 Relation between the limit of functions and its components

Proposition 5.2.

$$\lim_{x \rightarrow a} f(x) = \ell$$

if and only if

$$\lim_{x \rightarrow a} f_j(x) = \ell_j$$

for all $j = 1, \dots, m$.

Proof. See proposition 3.1, Chapter 2 from [Shi04]. □

Example 5.3. Fix a non-zero vector $b \in \mathbb{R}^n$. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be defined by $f(x) = b \cdot x$. Show that

$$\lim_{x \rightarrow a} f(x) = b \cdot a$$

Answer: Fix a nonzero vector $b \in \mathbb{R}^n$. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be defined by

$$f(x) = b \cdot x.$$

We claim that

$$\lim_{x \rightarrow a} f(x) = b \cdot a$$

because

$$|f(x) - b \cdot a| = |b \cdot x - b \cdot a| = |b \cdot (x - a)| \leq \|b\| \|x - a\|,$$

by the Cauchy-Schwarz Inequality. Thus, given $\varepsilon > 0$, if we take

$$\delta = \frac{\varepsilon}{\|b\|},$$

then whenever $0 < \|x - a\| < \delta$, we have

$$|f(x) - b \cdot a| < \|b\| \frac{\varepsilon}{\|b\|} = \varepsilon,$$

as needed.

Example 5.4. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be defined by $f(x) = \|x\|^2$. Show that

$$\lim_{x \rightarrow a} f(x) = \|a\|^2$$

Answer: We have

$$f(x) = \|x\|^2 = x \cdot x,$$

so

$$f(x) - \|a\|^2 = \|x\|^2 - \|a\|^2 = x \cdot x - a \cdot a.$$

Now

$$x \cdot x - a \cdot a = (x - a) \cdot (x + a),$$

and therefore

$$|f(x) - \|a\|^2| = |(x - a) \cdot (x + a)| \leq \|x - a\| \|x + a\|$$

by the Cauchy–Schwarz inequality.

If $\|x - a\| < 1$, then

$$\|x\| \leq \|x - a\| + \|a\| < 1 + \|a\|,$$

so

$$\|x + a\| \leq \|x\| + \|a\| < 1 + 2\|a\|.$$

Hence, whenever $\|x - a\| < 1$,

$$|f(x) - \|a\|^2| \leq (1 + 2\|a\|)\|x - a\|.$$

Given $\varepsilon > 0$, choose

$$\delta = \min \left\{ 1, \frac{\varepsilon}{1 + 2\|a\|} \right\}.$$

Then if $0 < \|x - a\| < \delta$, we have

$$|f(x) - \|a\|^2| \leq (1 + 2\|a\|)\|x - a\| < (1 + 2\|a\|)\delta \leq \varepsilon.$$

Thus,

$$\lim_{x \rightarrow a} f(x) = \|a\|^2.$$

Example 5.5. Let $f : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}$ be defined by $f \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \frac{x^2 y}{x^2 + y^2}$. Does $\lim_{x \rightarrow 0} f(x)$ exist?

Example 5.6. Let $f : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}$ be defined by $f \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \frac{x^2}{x^2 + y^2}$. Does $\lim_{x \rightarrow 0} f(x)$ exist?

Example 5.7. Let $f : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}$ be defined by $f \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \frac{xy}{x^2 + y^2}$. Does $\lim_{x \rightarrow 0} f(x)$ exist?

These examples are taken from the textbook [Shi04], Examples 3,4 from Chapter 2, section 3.

Proposition 5.8. Suppose $U \subset \mathbb{R}^n$ is open and $f : U \rightarrow \mathbb{R}^m$. Then f is continuous at $a \in U$

if and only if for every sequence $\{x_k\}$ of points in U converging to a , the sequence $\{f(x_k)\}$ converges to $f(a)$.

Proof. I have uploaded the recorded video of the proof of this proposition on YouTube (<https://www.youtube.com/watch?v=wCV83G1EAGwURL>) which contain the detailed proof of this proposition.

□

6 Lecture-06

6.1 Limits of Functions (Contd.)

Definition 6.1. For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, we write

$$\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) = L$$

if for every $\epsilon > 0$, there exists $\delta > 0$ such that for all $\vec{x} \in \mathbb{R}^n$, except possibly $\vec{x} = \vec{a}$,

$$\|\vec{x} - \vec{a}\| < \delta \implies \|f(\vec{x}) - L\| < \epsilon.$$

Example 6.2. Consider,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2} = 0$$

Proof. Let $\epsilon > 0$ be given. Choose $\delta = \epsilon/3$. For any $(x, y) \neq (0, 0)$ with $\sqrt{x^2 + y^2} < \delta$, we have

$$\left| \frac{3x^2y}{x^2 + y^2} \right| \leq \frac{3x^2|y|}{x^2 + y^2} \leq 3|y| \leq 3\sqrt{x^2 + y^2} < 3\delta = \epsilon.$$

Therefore, $\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2} = 0.$ □

Example 6.3. Consider,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{5x^2y^2}{x^2 + y^2} = 0$$

Proof. Let $\epsilon > 0$ be given. Choose $\delta = \sqrt{\epsilon/5}$. For any $(x, y) \neq (0, 0)$ with $\sqrt{x^2 + y^2} < \delta$, we have

$$\left| \frac{5x^2y^2}{x^2 + y^2} \right| = 5 \cdot \frac{x^2}{x^2 + y^2} \cdot y^2 \leq 5 \cdot 1 \cdot y^2 \leq 5(x^2 + y^2) < 5\delta^2 = \epsilon.$$

Therefore, $\lim_{(x,y) \rightarrow (0,0)} \frac{5x^2y^2}{x^2 + y^2} = 0.$ □

Example 6.4. Consider,

$$\lim_{(x,y) \rightarrow (2,2)} \frac{x^3 - y^3}{x^2 - y^2} = 3$$

Proof. Let $\epsilon > 0$ be given. Choose $\delta = \min \left\{ 1, \frac{\epsilon}{6} \right\}$. For any $(x, y) \neq (2, 2)$ with $\sqrt{(x-2)^2 + (y-2)^2} < \delta$ and $x \neq y$ (so that the expression is defined), write $u = x - 2$ and $v = y - 2$. Then $|u|, |v| < \delta$.

Since $\delta \leq 1$, we have $|u + v| \leq |u| + |v| < 2\delta \leq 2$, hence

$$|x + y| = |4 + u + v| \geq 4 - |u + v| > 4 - 2 = 2.$$

For $x \neq y$, the function simplifies:

$$\frac{x^3 - y^3}{x^2 - y^2} = \frac{x^2 + xy + y^2}{x + y}.$$

Now compute the difference:

$$\left| \frac{x^2 + xy + y^2}{x + y} - 3 \right| = \frac{|(x^2 + xy + y^2) - 3(x + y)|}{|x + y|}.$$

Substituting $x = 2 + u$, $y = 2 + v$ gives

$$x^2 + xy + y^2 - 3x - 3y = 3(u + v) + (u^2 + v^2 + uv).$$

Therefore,

$$\left| \frac{x^2 + xy + y^2}{x + y} - 3 \right| \leq \frac{3|u + v| + |u|^2 + |v|^2 + |u||v|}{|x + y|} < \frac{3 \cdot 2\delta + \delta^2 + \delta^2 + \delta^2}{2} = \frac{6\delta + 3\delta^2}{2} = 3\delta + \frac{3}{2}\delta^2.$$

Since $\delta \leq 1$, we have $\frac{3}{2}\delta^2 \leq \frac{3}{2}\delta$. Hence

$$\left| \frac{x^2 + xy + y^2}{x + y} - 3 \right| < 3\delta + \frac{3}{2}\delta \leq \frac{9}{2}\delta.$$

Because $\delta \leq \varepsilon/6$, it follows that $\frac{9}{2}\delta \leq \frac{9}{2} \cdot \frac{\varepsilon}{6} = \frac{3}{4}\varepsilon < \varepsilon$.

Thus, for all (x, y) with $0 < \sqrt{(x-2)^2 + (y-2)^2} < \delta$ and $x \neq y$,

$$\left| \frac{x^3 - y^3}{x^2 - y^2} - 3 \right| < \varepsilon.$$

Hence $\lim_{(x,y) \rightarrow (2,2)} \frac{x^3 - y^3}{x^2 - y^2} = 3$. □

Example 6.5. Consider,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2} = 1$$

Proof. Let $\varepsilon > 0$ be given. Since $\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$, there exists $\delta_0 > 0$ such that for all t with $0 < |t| < \delta_0$, we have $\left| \frac{\sin t}{t} - 1 \right| < \varepsilon$. Choose $\delta = \sqrt{\delta_0}$. For any $(x, y) \neq (0, 0)$ with $\sqrt{x^2 + y^2} < \delta$, set $t = x^2 + y^2 > 0$. Then $0 < t < \delta^2 = \delta_0$, so $\left| \frac{\sin t}{t} - 1 \right| < \varepsilon$. Hence $\left| \frac{\sin(x^2 + y^2)}{x^2 + y^2} - 1 \right| < \varepsilon$.

Therefore $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2} = 1$. □

Example 6.6. Consider,

$$\lim_{(x,y) \rightarrow (1,1)} \frac{2(x - y^2)}{x^2 - y^4} = 1$$

Proof. Let $\varepsilon > 0$ be given. Choose $\delta = \min \left\{ \frac{1}{4}, \frac{\varepsilon}{4} \right\}$. Suppose (x, y) satisfies $0 < \sqrt{(x-1)^2 + (y-1)^2} < \delta$ and $x \neq y^2$ (so that the expression is defined). Then $|x - 1| < \delta$ and $|y - 1| < \delta$.

We estimate

$$|x + y^2 - 2| = |(x-1) + (y^2-1)| \leq |x-1| + |y-1||y+1| < \delta + \delta(|y-1| + 2) < \delta + \delta(\delta + 2) = \delta(\delta + 3).$$

Since $\delta \leq \frac{1}{4}$, we have $\delta + 3 \leq \frac{1}{4} + 3 = \frac{13}{4} < 4$, thus $|x + y^2 - 2| < 4\delta$. Moreover, $x + y^2 \geq 2 - |x + y^2 - 2| > 2 - 4\delta \geq 2 - 4 \cdot \frac{1}{4} = 1$, so $x + y^2 > 1$.

For $x \neq y^2$ we can simplify the original fraction:

$$\frac{2(x - y^2)}{x^2 - y^4} = \frac{2(x - y^2)}{(x - y^2)(x + y^2)} = \frac{2}{x + y^2}.$$

Hence

$$\left| \frac{2(x - y^2)}{x^2 - y^4} - 1 \right| = \left| \frac{2}{x + y^2} - 1 \right| = \frac{|2 - (x + y^2)|}{x + y^2} < \frac{4\delta}{1} = 4\delta \leq \varepsilon.$$

Therefore, $\lim_{(x,y) \rightarrow (1,1)} \frac{2(x - y^2)}{x^2 - y^4} = 1.$

□

7 Lecture-07

From our first lecture, we have seen the general definition of derivative of a function.

Definition 7.1. Let U be an open set of $\mathbb{R}$ and $f : U \rightarrow \mathbb{R}$ then f is differentiable at a with the derivative $f'(a)$ if and only if

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - f'(a)h}{h} = 0$$

7.1 First order derivatives

In multivariable calculus, we understand a function by knowing how it varies with each variable, fixing all the others. We begin by considering a real-valued function f of two variables x and y .

Definition 7.2. We define the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ as

$$\frac{\partial f}{\partial x} \begin{pmatrix} a \\ b \end{pmatrix} = \lim_{h \rightarrow 0} \frac{f \begin{pmatrix} a+h \\ b \end{pmatrix} - f \begin{pmatrix} a \\ b \end{pmatrix}}{h}$$

and

$$\frac{\partial f}{\partial y} \begin{pmatrix} a \\ b \end{pmatrix} = \lim_{k \rightarrow 0} \frac{f \begin{pmatrix} a \\ b+k \end{pmatrix} - f \begin{pmatrix} a \\ b \end{pmatrix}}{k}$$

respectively, provided these limits exist.

For get a clear understanding have a look the figure 1.1 from [Shi04], chapter 3, section 1.

Example 7.3. Given that $f(x, y) = x^2 + 2y$, find $\frac{\partial f}{\partial x} \begin{pmatrix} a \\ b \end{pmatrix}$ and $\frac{\partial f}{\partial y} \begin{pmatrix} a \\ b \end{pmatrix}$.

Answer: We have shown it in the classroom.

More generally, if $U \subset \mathbb{R}^n$ is open and $a \in U$, we define the j^{th} partial derivative of

$$f : U \rightarrow \mathbb{R}^m$$

at a to be

$$\frac{\partial f}{\partial x_j}(a) = \lim_{t \rightarrow 0} \frac{f(a + te_j) - f(a)}{t}, \quad j = 1, \dots, n$$

(provided this limit exists). Many authors use the alternative notation $D_j f(a)$ to represent the j^{th} partial derivative of f at a .

The partial derivatives of f measure the rate of change of f in the directions of the coordinate axes, i.e., in the directions of the standard basis vectors $e_1, \dots, e_n$. Given any nonzero vector $\mathbf{v}$, it is natural to consider the rate of change of f in the direction of $\mathbf{v}$.

Definition 7.4. Let $U \subset \mathbb{R}^n$ be open and $a \in U$. We define the *directional derivative* of

$$f : U \rightarrow \mathbb{R}^m$$

at a in the direction $\mathbf{v}$ to be

$$D_{\mathbf{v}}f(a) = \lim_{t \rightarrow 0} \frac{f(a + t\mathbf{v}) - f(a)}{t},$$

provided this limit exists.

Example 7.5. Let

$$f \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x^2y + e^{3x+y-z}, \quad \text{let } a = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}.$$

What is the directional derivative $D_{\mathbf{v}}f(a)$?

Approach-01: We have shown it in the classroom using the definition.

Approach-02: We define $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ by $\varphi(t) = f(a + t\mathbf{v})$. Note that

$$\varphi'(0) = \lim_{t \rightarrow 0} \frac{\varphi(t) - \varphi(0)}{t} = \lim_{t \rightarrow 0} \frac{f(a + t\mathbf{v}) - f(a)}{t} = D_{\mathbf{v}}f(a).$$

So we just calculate φ and compute its derivative at 0:

$$\begin{aligned} \varphi(t) &= f \begin{pmatrix} 1 + 2t \\ -1 + 3t \\ 2 - t \end{pmatrix} \\ &= (1 + 2t)^2(-1 + 3t) + e^{3(1+2t)+(-1+3t)-(2-t)} \\ &= (1 + 2t)^2(-1 + 3t) + e^{10t} \end{aligned}$$

Hence,

$$\varphi'(t) = 4(1 + 2t)(-1 + 3t) + 3(1 + 2t)^2 + 10e^{10t},$$

from which we conclude that

$$D_{\mathbf{v}}f(a) = \varphi'(0) = 9.$$

This approach is usually more convenient than calculating the limit directly when we want skip the tedious limit definition.

7.2 Higher Order Partial Derivatives

Given a function $f : U \rightarrow \mathbb{R}$ where $U \subset \mathbb{R}^n$ is open, we can consider the partial derivatives of f . If these partial derivatives are themselves differentiable, we can take their partial derivatives, which are called the second order partial derivatives of f . We can continue this process to define higher order partial derivatives.

For a function $f : U \rightarrow \mathbb{R}$ where $U \subset \mathbb{R}^n$ is open, we denote the second order partial derivatives of f by

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right), \quad i, j = 1, \dots, n.$$

The second order partial derivatives of f are often denoted by

$$f_{x_i x_j} = \frac{\partial^2 f}{\partial x_i \partial x_j}.$$

Example 7.6. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = x^2y + e^{3x+y}.$$

Find the second order partial derivatives of f .

Example 7.7. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{xy(x^2-y^2)}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}.$$

Calculate the mixed partial derivatives $f_{xy} = \frac{\partial^2 f}{\partial x \partial y}$ and $f_{yx} = \frac{\partial^2 f}{\partial y \partial x}$ at the origin. And show that $f_{xy}(0, 0) \neq f_{yx}(0, 0)$.

Answer: We have shown it in the classroom.

The most surprising fact about second order partial derivatives is that, under mild conditions, the order of differentiation does not matter. In other words, if $f : U \rightarrow \mathbb{R}$ is a function where $U \subset \mathbb{R}^n$ is open, then

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}, \quad i, j = 1, \dots, n,$$

provided the second order partial derivatives are continuous at the point in question. This result is known as *Clairaut's theorem* or *Schwarz's theorem*.

8 Lecture-08

8.1 Directional Derivative (contd.)

Problem 8.1. If $T(\vec{x})$ is a linear transformation, then find the directional derivative of $T(\vec{x})$ at $\vec{x} = \vec{a}$ in the direction $\vec{v}$.

Problem 8.2. If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is defined by $f(\vec{x}) = \|\vec{x}\|^2$, find the directional derivative of f at $\vec{x} = \vec{a}$ in the direction $\vec{v}$.

Solution:

$$\begin{aligned} D_{\vec{v}}f(\vec{a}) &= \lim_{t \rightarrow 0} \frac{f(\vec{a} + t\vec{v}) - f(\vec{a})}{t} \\ &= \lim_{t \rightarrow 0} \frac{\|\vec{a} + t\vec{v}\|^2 - \|\vec{a}\|^2}{t} \\ &= \lim_{t \rightarrow 0} \frac{(\vec{a} + t\vec{v}) \cdot (\vec{a} + t\vec{v}) - \vec{a} \cdot \vec{a}}{t} \\ &= \lim_{t \rightarrow 0} \frac{2t\vec{a} \cdot \vec{v} + t^2\|\vec{v}\|^2}{t} \\ &= 2\vec{a} \cdot \vec{v}. \end{aligned}$$

Example 8.3. Let

$$f(x, y) = \begin{cases} \frac{x^3}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0), \end{cases}$$

Find the partial derivatives and directional derivatives of f at $(0, 0)$.

Partial derivatives at $(0, 0)$:

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{h^3}{h^2} - 0}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1.$$

$$f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k) - f(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{0 - 0}{k} = 0.$$

Directional derivative at $(0, 0)$: Let $\vec{v} = (a, b) \neq (0, 0)$. Then

$$D_{\vec{v}}f(0, 0) = \lim_{t \rightarrow 0} \frac{f(ta, tb) - f(0, 0)}{t} = \lim_{t \rightarrow 0} \frac{\frac{(ta)^3}{(ta)^2 + (tb)^2}}{t} = \lim_{t \rightarrow 0} \frac{\frac{t^3 a^3}{t^2(a^2 + b^2)}}{t} = \frac{a^3}{a^2 + b^2}.$$

Equivalently, if $\vec{u} = (u_1, u_2)$ is a unit direction, then

$$D_{\vec{u}}f(0, 0) = \frac{u_1^3}{u_1^2 + u_2^2} = u_1^3 \neq 0.$$

But still f is not differentiable at $(0, 0)$, see <https://math.stackexchange.com/a/3057145/1167703>.

Question: Does the existence of all directional derivatives at a point imply the existence of the total derivative at that point? The answer is no, as shown by the above example. The

function f has all directional derivatives at $(0,0)$, but it is not differentiable at $(0,0)$ because the limit in the definition of differentiability does not exist.

8.2 Total Derivative

Definition 8.4. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called *differentiable at $x = a$* if there exists a real number L such that

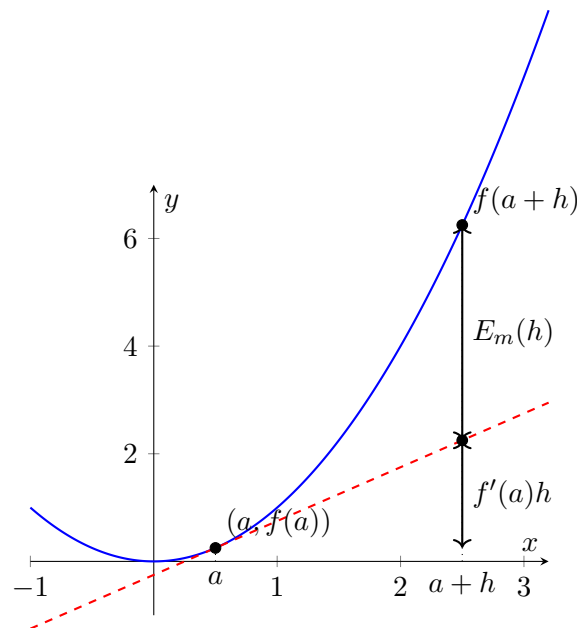
$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - Lh}{h} = 0.$$

Then the number L is called the *derivative* of f at $x = a$.

The definition indicates that,

$$f(a+h) = f(a) + Lh + o(h)$$

So, near $x = a$, the function f behaves like a linear function with slope L , $f(a+h) \approx f(a) + Lh = \text{base point} + \text{linear correction}$. The derivative only captures the linear correction part. The constant term just shifts the graph vertically.



Remark: The notation $o(h)$ means that the error term $E(h) = f(a+h) - f(a) - Lh$ is small compared to h as $h \rightarrow 0$. More precisely, it means that $\lim_{h \rightarrow 0} E(h)/h = 0$. In other words, $E(h)$ goes to zero faster than h does. Formally, we say a function $E(h)$ is $o(h)$ if

$$\lim_{h \rightarrow 0} E(h)/h = 0.$$

In one variable $h^2 = o(h)$, $h^3 = o(h)$, but $2h$ is not $o(h)$ because $\lim_{h \rightarrow 0} 2h/h = 2 \neq 0$.

Definition 8.5. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called *differentiable at $x = \vec{a}$* if there exists a linear transformation $T_{\vec{a}} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$\lim_{\vec{h} \rightarrow 0} \frac{\|f(\vec{a} + \vec{h}) - f(\vec{a}) - T_{\vec{a}}(\vec{h})\|}{\|\vec{h}\|} = 0.$$

Then $T_{\vec{a}}$ is called the *total derivative* at $x = \vec{a}$, and is often denoted by $Df(\vec{a})$.

Again, this says that $Df(\vec{a})$ is the best linear approximation to the function $f - f(\vec{a})$ at $\vec{a}$, in the sense that the difference $f(\vec{a} + \vec{h}) - f(\vec{a}) - Df(\vec{a})(\vec{h})$ is small compared to $\|\vec{h}\|$ as $\vec{h} \rightarrow 0$.

Proposition 8.6. *Suppose there exist two linear transformations*

$$T_{\vec{a}}^{(1)}, T_{\vec{a}}^{(2)} : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

such that

$$\lim_{\|\vec{h}\| \rightarrow 0} \frac{\|f(\vec{a} + \vec{h}) - f(\vec{a}) - T_{\vec{a}}^{(j)}(\vec{h})\|}{\|\vec{h}\|} = 0, \quad j = 1, 2.$$

Prove that

$$T_{\vec{a}}^{(1)} = T_{\vec{a}}^{(2)}.$$

Proof. Since both limits are zero,

$$T_{\vec{a}}^{(2)}(\vec{h}) - T_{\vec{a}}^{(1)}(\vec{h}) = \{f(\vec{a} + \vec{h}) - f(\vec{a}) - T_{\vec{a}}^{(1)}(\vec{h})\} - \{f(\vec{a} + \vec{h}) - f(\vec{a}) - T_{\vec{a}}^{(2)}(\vec{h})\}.$$

Taking norms and applying the triangle inequality gives

$$\|(T_{\vec{a}}^{(2)} - T_{\vec{a}}^{(1)})(\vec{h})\| \leq \|f(\vec{a} + \vec{h}) - f(\vec{a}) - T_{\vec{a}}^{(1)}(\vec{h})\| + \|f(\vec{a} + \vec{h}) - f(\vec{a}) - T_{\vec{a}}^{(2)}(\vec{h})\|.$$

Dividing by $\|\vec{h}\|$ and taking the limit yields

$$\lim_{\|\vec{h}\| \rightarrow 0} \frac{\|(T_{\vec{a}}^{(2)} - T_{\vec{a}}^{(1)})(\vec{h})\|}{\|\vec{h}\|} = 0.$$

Let

$$L := T_{\vec{a}}^{(2)} - T_{\vec{a}}^{(1)}.$$

Then L is linear and

$$\lim_{\|\vec{h}\| \rightarrow 0} \frac{\|L(\vec{h})\|}{\|\vec{h}\|} = 0.$$

This only tells us something about vectors $\vec{h}$ near $\vec{0}$. It does not immediately tell you what $L(\vec{h})$ is for a fixed nonzero $\vec{h}$. Uniqueness needs a statement like the operator is zero on every vector. The trick plug in $\alpha\vec{h}$ where $\alpha \rightarrow 0$ is exactly how you turn a local limit statement into a global conclusion using linearity.

Let $\vec{h} \neq 0$ be fixed. For any scalar $\alpha \rightarrow 0$, we have $\alpha\vec{h} \rightarrow 0$. Thus the limit applies:

$$0 = \lim_{\alpha \rightarrow 0} \frac{\|L(\alpha\vec{h})\|}{\|\alpha\vec{h}\|}.$$

By linearity,

$$L(\alpha\vec{h}) = \alpha L(\vec{h}),$$

so

$$\frac{\|L(\alpha\vec{h})\|}{\|\alpha\vec{h}\|} = \frac{\|\alpha L(\vec{h})\|}{|\alpha| \|\vec{h}\|} = \frac{\|L(\vec{h})\|}{\|\vec{h}\|}.$$

$$0 = \lim_{\alpha \rightarrow 0} \frac{\|L(\alpha \vec{h})\|}{\|\alpha \vec{h}\|} = \lim_{\alpha \rightarrow 0} \frac{\|L(\vec{h})\|}{\|\vec{h}\|}.$$

Notice the quantity inside the limit is constant in α . The limit of a constant sequence/function is just the constant itself. So

$$\frac{\|L(\vec{h})\|}{\|\vec{h}\|} = 0 \implies \|L(\vec{h})\| = 0.$$

And the only vector with zero norm is the zero vector. Hence $L(\vec{h}) = 0$. Since the above holds for every nonzero vector $\vec{h}$, $L \equiv 0$. Thus

$$T_{\vec{a}}^{(2)} = T_{\vec{a}}^{(1)}.$$

□

Example 8.7. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = x^2 - y^2$$

Find the derivative of f at the point $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$, find the value of $Df \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

Solution. Let

$$Df \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} a & b \end{pmatrix}$$

Using the definition of differentiability, let $h = (h_1, h_2) \rightarrow (0, 0)$. Then

$$\begin{aligned} & \lim_{\|\vec{h}\| \rightarrow 0} \frac{\left\| f \left(\begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} \right) - f \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} \right\|}{\|\vec{h}\|} \\ &= \lim_{\|\vec{h}\| \rightarrow 0} \frac{\left\| f \begin{pmatrix} 2+h_1 \\ 3+h_2 \end{pmatrix} - f \begin{pmatrix} 2 \\ 3 \end{pmatrix} - ah_1 - bh_2 \right\|}{\sqrt{h_1^2 + h_2^2}}. \end{aligned}$$

Compute:

$$f(2+h_1, 3+h_2) - f(2, 3) = (2+h_1)^2 - (3+h_2)^2 - (2^2 - 3^2).$$

and

$$(2+h_1)^2 - (3+h_2)^2 - 2^2 + 3^2 = (4 + 4h_1 + h_1^2) - (9 + 6h_2 + h_2^2) - 4 + 9 = 4h_1 - 6h_2 + h_1^2 - h_2^2.$$

So the numerator becomes

$$|(4-a)h_1 - (6+b)h_2 + h_1^2 - h_2^2|.$$

To make the linear terms vanish (**why?**), choose

$$4-a=0 \Rightarrow a=4, \quad -(6+b)=0 \Rightarrow b=-6.$$

Then the limit reduces to

$$\lim_{(h_1, h_2) \rightarrow (0,0)} \frac{|h_1^2 - h_2^2|}{\sqrt{h_1^2 + h_2^2}} \leq \lim_{(h_1, h_2) \rightarrow (0,0)} \frac{h_1^2 + h_2^2}{\sqrt{h_1^2 + h_2^2}} = \lim_{(h_1, h_2) \rightarrow (0,0)} \sqrt{h_1^2 + h_2^2} = 0.$$

Hence f is differentiable at $(2, 3)$ and

$$Df \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 & -6 \end{pmatrix}.$$

Example 8.8. For $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$f(x, y) = \begin{pmatrix} xy \\ y^2 \end{pmatrix},$$

find the value of $Df \begin{pmatrix} a \\ b \end{pmatrix}$.

Solution: For $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$f(x, y) = \begin{pmatrix} xy \\ y^2 \end{pmatrix},$$

the derivative at a point $\vec{a} = \begin{pmatrix} a \\ b \end{pmatrix}$ is the linear map $Df \begin{pmatrix} a \\ b \end{pmatrix}$ satisfying

$$f \begin{pmatrix} a+h \\ b+k \end{pmatrix} - f \begin{pmatrix} a \\ b \end{pmatrix} = Df_{\vec{a}} \begin{pmatrix} h \\ k \end{pmatrix} + o(\|(h, k)\|).$$

We will see this definition can be used to compute the derivative without using the limit definition directly. Compute the increment:

$$f \begin{pmatrix} a+h \\ b+k \end{pmatrix} = \begin{pmatrix} (a+h)(b+k) \\ (b+k)^2 \end{pmatrix} = \begin{pmatrix} ab + ak + bh + hk \\ b^2 + 2bk + k^2 \end{pmatrix}.$$

So

$$f \begin{pmatrix} a+h \\ b+k \end{pmatrix} - f \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ak + bh + hk \\ 2bk + k^2 \end{pmatrix}.$$

The linear part in (h, k) is

$$\begin{pmatrix} bh + ak \\ 2bk \end{pmatrix},$$

and the remainder is

$$\begin{pmatrix} hk \\ k^2 \end{pmatrix},$$

(you can skip to verify the error term is $o(\|(h, k)\|)$, but we will verify it for the sake of completeness) whose norm is

which is $o(\|(h, k)\|)$ because $|hk| \leq \frac{1}{2}(h^2 + k^2)$ and $|k^2| \leq h^2 + k^2$.

Claim: $\sqrt{x^2 + y^2} \leq |x| + |y|$.

Proof: $(|x| + |y|)^2 = x^2 + y^2 + 2|x||y| \geq x^2 + y^2$.

Since both sides are nonnegative, taking square roots gives

$$\sqrt{x^2 + y^2} \leq |x| + |y|.$$

$$\|R(h, k)\| = \sqrt{(hk)^2 + (k^2)^2} \leq |hk| + |k^2| \leq \frac{h^2 + k^2}{2} + (h^2 + k^2) = \frac{3}{2}(h^2 + k^2) = \frac{3}{2}\|(h, k)\|^2.$$

$$\frac{\|R(h, k)\|}{\|(h, k)\|} \leq \frac{3}{2}\|(h, k)\| \rightarrow 0,$$

Therefore,

$$Df_{\vec{a}} \begin{pmatrix} h \\ k \end{pmatrix} = \begin{pmatrix} bh + ak \\ 2bk \end{pmatrix},$$

So the matrix of $Df_{\vec{a}}$ is

$$Df_{\vec{a}} = \begin{pmatrix} b & a \\ 0 & 2b \end{pmatrix}.$$

Remark 8.9. For more interesting example related with matrix algebra can be found in Hubbard [HH02] (Example 1.7.17 and proposition 1.7.18).

8.3 Tangent Space for $\mathbb{R}^2$ (Optional)

For a *fixed* direction v , the directional derivative at p defines a map

$$D_v|_p : C^\infty(\mathbb{R}^2) \longrightarrow \mathbb{R}, \quad f \longmapsto D_v f(p),$$

which is linear. Moreover, it satisfies the Leibniz rule

$$D_v(fg)(p) = f(p) D_v g(p) + g(p) D_v f(p),$$

so $D_v|_p$ is a *derivation at p* (the “tangent vector as operator” viewpoint).

Definition 8.10. A derivation is a linear map

$$X : C^\infty(\mathbb{R}^2) \rightarrow \mathbb{R}$$

acting on smooth functions, satisfying the Leibniz rule:

$$X(fg) = X(f)g + fX(g).$$

It generalizes the concept of a tangent vector as an operator, mapping a function to its directional derivative at a specific point or over a region.

The tangent space is *not* built from one fixed operator $D_v|_p$; rather it is built from the

collection of *all* such derivations at p . A clean definition is

$$T_p\mathbb{R}^2 := \left\{ X : C^\infty(\mathbb{R}^2) \rightarrow \mathbb{R} \mid X \text{ is linear and } X(fg) = f(p)X(g) + g(p)X(f) \right\}.$$

Elements of $T_p\mathbb{R}^2$ are operators that differentiate functions at p . Each direction $v = (v_1, v_2) \in \mathbb{R}^2$ produces an element

$$v \mapsto D_v|_p \in T_p\mathbb{R}^2.$$

Define the standard derivations

$$\frac{\partial}{\partial x}\bigg|_p(f) = \frac{\partial f}{\partial x}(p), \quad \frac{\partial}{\partial y}\bigg|_p(f) = \frac{\partial f}{\partial y}(p).$$

Then for $v = (v_1, v_2)$,

$$D_v|_p = v_1 \frac{\partial}{\partial x}\bigg|_p + v_2 \frac{\partial}{\partial y}\bigg|_p,$$

so $\left\{ \frac{\partial}{\partial x}\bigg|_p, \frac{\partial}{\partial y}\bigg|_p \right\}$ is a basis of $T_p\mathbb{R}^2$. A tangent vector at p can be viewed as an operator that takes a smooth function f and returns its directional rate of change at p . The tangent space $T_p\mathbb{R}^2$ is the vector space of all such derivative-operators (derivations), and in coordinates it has basis $\frac{\partial}{\partial x}\big|_p, \frac{\partial}{\partial y}\big|_p$. See <https://math.stackexchange.com/q/1118419/1167703> to get idea why are derivations useful for defining tangent vectors.

Example 8.11. Let $f(x, y) = 3x - 2y$ and $p = (1, 1)$. For $v = (1, 0)$ and $w = (0, 1)$,

$$D_v|_p(f) = \frac{\partial f}{\partial x}(p) = 3, \quad D_w|_p(f) = \frac{\partial f}{\partial y}(p) = -2.$$

For $u = (2, 5)$,

$$D_u|_p(f) = 2 \cdot 3 + 5 \cdot (-2) = 6 - 10 = -4.$$

Equivalently,

$$D_u|_p = 2 \frac{\partial}{\partial x}\bigg|_p + 5 \frac{\partial}{\partial y}\bigg|_p.$$

Example 8.12. Let $f(x, y) = x^2 + y^2$ and $p = (1, 2)$. Then

$$f_x = 2x, \quad f_y = 2y \quad \Rightarrow \quad f_x(p) = 2, \quad f_y(p) = 4.$$

For $v = (3, -1)$,

$$D_v|_p(f) = 3 \cdot 2 + (-1) \cdot 4 = 6 - 4 = 2, \quad \text{hence} \quad D_{(3, -1)}|_p = 3 \frac{\partial}{\partial x}\bigg|_p - \frac{\partial}{\partial y}\bigg|_p.$$

Example 8.13. From the previous example, for $v = (1, 0)$ and $w = (0, 1)$,

$$D_v|_p(f) = 2, \quad D_w|_p(f) = 4.$$

Since $v + w = (1, 1)$,

$$D_{v+w}|_p(f) = 1 \cdot 2 + 1 \cdot 4 = 6, \quad \text{and indeed} \quad D_{v+w}|_p(f) = D_v|_p(f) + D_w|_p(f).$$

Example 8.14. Let $f(x, y) = x$, $g(x, y) = y^2$, $p = (1, 2)$, and $v = (3, -1)$. Then

$$D_v|_p(f) = 3, \quad D_v|_p(g) = (-1) \cdot (2y)|_{(1,2)} = -4.$$

For the product $fg = xy^2$, compute

$$(xy^2)_x = y^2, \quad (xy^2)_y = 2xy,$$

so at $(1, 2)$ we have $D(fg)(1, 2) = (4, 4)$ and hence

$$D_v|_p(fg) = (4, 4) \cdot (3, -1) = 12 - 4 = 8.$$

On the other hand,

$$f(p) D_v|_p(g) + g(p) D_v|_p(f) = (1)(-4) + (4)(3) = -4 + 12 = 8,$$

Tangent spaces let us define:

1. Derivatives
2. Gradients
3. Vector fields
4. Flows
5. Differential equations

9 Lecture-09

Proposition 9.1. *If the function $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at $\vec{a}$, then the partial derivatives $\frac{\partial f}{\partial x_i}(\vec{a})$ exists for all $i = 1, 2, \dots, n$ and $Df(\vec{a})$ is given by the row vector*

$$Df(\vec{a}) = \left(\frac{\partial f}{\partial x_1}(\vec{a}) \quad \frac{\partial f}{\partial x_2}(\vec{a}) \quad \cdots \quad \frac{\partial f}{\partial x_n}(\vec{a}) \right).$$

Proof. f is differentiable at $\vec{a}$ means that there exists a linear transformation $L \equiv Df(\vec{a}) : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$f(\vec{a} + \vec{h}) = f(\vec{a}) + L(\vec{h}) + o(\|\vec{h}\|).$$

where $\lim_{\|\vec{h}\| \rightarrow 0} \frac{o(\|\vec{h}\|)}{\|\vec{h}\|} = 0$. Suppose,

$$L(\vec{h}) = L \begin{pmatrix} h_1 \\ h_2 \\ \vdots \\ h_n \end{pmatrix} = \begin{pmatrix} c_1 & c_2 & \cdots & c_n \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \\ \vdots \\ h_n \end{pmatrix} = c_1 h_1 + c_2 h_2 + \cdots + c_n h_n.$$

Now, we want to impose the definition in each of the coordinate directions. For $i = 1, 2, \dots, n$, let $\vec{h} = t\vec{e}_i$ where $\vec{e}_i$ is the i -th standard basis vector. Then

$$\begin{aligned} f(\vec{a} + t\vec{e}_i) &= f(\vec{a}) + L(t\vec{e}_i) + o(\|t\vec{e}_i\|) \\ f(\vec{a} + t\vec{e}_i) - f(\vec{a}) &= tL(\vec{e}_i) + o(|t|) \\ \frac{f(\vec{a} + t\vec{e}_i) - f(\vec{a})}{t} &= L(\vec{e}_i) + \frac{o(|t|)}{t} \\ \lim_{t \rightarrow 0} \frac{f(\vec{a} + t\vec{e}_i) - f(\vec{a})}{t} &= L(\vec{e}_i) + \lim_{t \rightarrow 0} \frac{o(|t|)}{t} \\ \frac{\partial f}{\partial x_i}(\vec{a}) &= L(\vec{e}_i) = c_i. \end{aligned}$$

□

Proposition 9.2. *If the function $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at $\vec{a}$, then the partial derivatives $\frac{\partial f^j}{\partial x_i}(\vec{a})$ exists for all $i = 1, 2, \dots, n$, $j = 1, 2, \dots, m$ and $Df(\vec{a})$ is given by the matrix*

$$Df(\vec{a}) = \begin{pmatrix} \frac{\partial f^1}{\partial x_1}(\vec{a}) & \frac{\partial f^1}{\partial x_2}(\vec{a}) & \cdots & \frac{\partial f^1}{\partial x_n}(\vec{a}) \\ \frac{\partial f^2}{\partial x_1}(\vec{a}) & \frac{\partial f^2}{\partial x_2}(\vec{a}) & \cdots & \frac{\partial f^2}{\partial x_n}(\vec{a}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f^m}{\partial x_1}(\vec{a}) & \frac{\partial f^m}{\partial x_2}(\vec{a}) & \cdots & \frac{\partial f^m}{\partial x_n}(\vec{a}) \end{pmatrix}.$$

Proposition 9.3. (Optional) *Let $f : \mathbb{R}^k \rightarrow \mathbb{R}^n$ be a differentiable map. For each $i \in \{1, \dots, n\}$ let*

$$\pi^i : \mathbb{R}^n \rightarrow \mathbb{R}, \quad \pi^i(a^1, \dots, a^n) = a^i$$

be the i -th coordinate projection, and define the i -th component function of f by

$$f^i := \pi^i \circ f : \mathbb{R}^k \rightarrow \mathbb{R}.$$

Then for every $a \in \mathbb{R}^k$,

$$D(f^i)(a) = \pi^i \circ Df(a).$$

Equivalently, as linear maps $\mathbb{R}^k \rightarrow \mathbb{R}$,

$$D(f^i)(a)[h] = \pi^i(Df(a)[h]) \quad \text{for all } h \in \mathbb{R}^k.$$

Proof. See the proof from <https://math.stackexchange.com/a/4214572/1167703>. $\square$

Proposition 9.4. If $f : A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at x_0 , then f is continuous at x_0 .

Proof. By definition of the derivative of f we know that the limit

$$\lim_{h \rightarrow 0} \frac{\|f(x_0 + h) - f(x_0) + Dh\|}{\|h\|} = 0,$$

for some linear transformation D . Therefore since this limit exists, the following is true:

$$0 = \left(\lim_{h \rightarrow 0} \|h\| \right) \left(\lim_{h \rightarrow 0} \frac{\|f(x_0 + h) - f(x_0) + Dh\|}{\|h\|} \right) = \lim_{h \rightarrow 0} \|f(x_0 + h) - f(x_0) + Dh\|.$$

By the triangle inequality,

$$\lim_{h \rightarrow 0} \|f(x_0 + h) - f(x_0)\| \leq \lim_{h \rightarrow 0} \|f(x_0 + h) - f(x_0) + Dh\| + \lim_{h \rightarrow 0} \|Dh\| = \lim_{h \rightarrow 0} \|Dh\|.$$

But since D is linear, it is continuous and $D0 = 0$, so the lattermost limit is just 0. Thus 0 is an upper bound for the limit, but since the norm is always non-negative, this implies that the limit is in fact 0. $\square$

Example 9.5.

$$f(x, y, z) = x^2 + y^2 + z^2.$$

Take

$$\vec{a} = (1, 1, 1).$$

$$Df(x, y, z) = (2x, 2y, 2z).$$

$$Df(1, 1, 1) = (2, 2, 2).$$

Thus, for $\vec{v} = (v_1, v_2, v_3)$,

$$Df(1, 1, 1)(\vec{v}) = 2v_1 + 2v_2 + 2v_3.$$

$$D_{\vec{v}}f(1, 1, 1) = \lim_{t \rightarrow 0} \frac{(1 + tv_1)^2 + (1 + tv_2)^2 + (1 + tv_3)^2 - 3}{t}.$$

Expanding,

$$(1 + tv_1)^2 + (1 + tv_2)^2 + (1 + tv_3)^2 = 3 + 2t(v_1 + v_2 + v_3) + O(t^2).$$

Therefore,

$$D_{\vec{v}}f(1, 1, 1) = \lim_{t \rightarrow 0} \frac{3 + 2t(v_1 + v_2 + v_3) + O(t^2) - 3}{t} = 2(v_1 + v_2 + v_3).$$

Hence,

$$\boxed{D_{\vec{v}}f(\vec{a}) = Df(\vec{a})(\vec{v})}$$

We can see why it works in general. If f is differentiable at $\vec{a}$, then

$$\begin{aligned} f(\vec{a} + \vec{h}) &= f(\vec{a}) + Df(\vec{a})(\vec{h}) + o(|\vec{h}|) \\ \xrightarrow{\vec{h}=t\vec{v}} \frac{f(\vec{a} + t\vec{v}) - f(\vec{a})}{t} &= Df(\vec{a})(\vec{v}) + \frac{o(t|\vec{v}|)}{t}. \end{aligned}$$

As $t \rightarrow 0$, the error term goes to 0, giving

$$D_{\vec{v}}f(\vec{a}) = Df(\vec{a})(\vec{v}).$$

For more rigorous proof, see the following proposition.

Proposition 9.6. *When f is differentiable at $\vec{a}$, for any $\vec{v}$, the directional derivative of f at $\vec{a}$ in the direction $\vec{v}$ exists and is given by*

$$D_{\vec{v}}f(\vec{a}) = Df(\vec{a})(\vec{v}).$$

Proof. See the proof from Theodore Shifrin [Shi04] (Chapter 3, section 2, proposition 2.3). $\square$

Example 9.7. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by,

$$f \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} xy \\ x^2 - y^2 \end{pmatrix}$$

At the point $\vec{a} = (1, 1)$, at what rate f varying in the direction $\vec{v} = (1, 0)$?

The derivative of f at $\vec{a}$ is,

$$Df(\vec{x}) = \begin{pmatrix} y & x \\ 2x & -2y \end{pmatrix}.$$

Thus,

$$D_{\vec{v}}f(\vec{a}) = Df(\vec{a})(\vec{v}) = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

See, how the directional derivative is just the application of the total derivative to the direction vector.

Proposition 9.8. *If $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ has continuous partial derivatives, then f is differentiable.*

Proof. See the proof from Theodore Shifrin [Shi04] (Chapter 3, section 2, proposition 2.4). $\square$

9.1 Jacobian Matrix

Definition 9.9. For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, written as

$$f(\vec{x}) = f \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} f_1(x_1, \dots, x_n) \\ f_2(x_1, \dots, x_n) \\ \vdots \\ f_m(x_1, \dots, x_n) \end{pmatrix} = \begin{pmatrix} f_1(\vec{x}) \\ f_2(\vec{x}) \\ \vdots \\ f_m(\vec{x}) \end{pmatrix},$$

the Jacobian matrix of f at a point $x \in \mathbb{R}^n$ is the $(m \times n)$ matrix of first-order partial derivatives:

$$J_f(x) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(x) & \frac{\partial f_1}{\partial x_2}(x) & \cdots & \frac{\partial f_1}{\partial x_n}(x) \\ \frac{\partial f_2}{\partial x_1}(x) & \frac{\partial f_2}{\partial x_2}(x) & \cdots & \frac{\partial f_2}{\partial x_n}(x) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(x) & \frac{\partial f_m}{\partial x_2}(x) & \cdots & \frac{\partial f_m}{\partial x_n}(x) \end{bmatrix}.$$

Its (i, j) -entry is

$$(J_f(x))_{ij} = \frac{\partial f_i}{\partial x_j}(x).$$

It represents the best linear approximation of f near x . If $m = 1$, the Jacobian is just the row vector of partial derivatives, closely related to the gradient.

Theorem 9.10. *If f is differentiable at $\vec{a}$, then all the partial derivatives of f at $\vec{a}$ exist and the matrix representing $Df(\vec{a})$ is the Jacobian matrix $J_f(\vec{a})$.*

Proof. We already proved this in Proposition 9.2. □

Remark 9.11. The gradient of a scalar-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the row vector of partial derivatives, which is the Jacobian matrix in this case.

$$\text{grad } f(\vec{a}) = \nabla f(\vec{a}) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(\vec{a}) \\ \frac{\partial f}{\partial x_2}(\vec{a}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\vec{a}) \end{pmatrix}.$$

In other words, $\text{grad } f(\vec{a}) = [Df(\vec{a})]^T = \left(\frac{\partial f}{\partial x_1}(\vec{a}) \quad \frac{\partial f}{\partial x_2}(\vec{a}) \quad \cdots \quad \frac{\partial f}{\partial x_n}(\vec{a}) \right)^T$. The transpose is not as innocent as it looks. geometrically, it is only defined if we are in a space with an inner product. When there is no nature inner product, the vector $\vec{\nabla} f(\vec{a})$ is not natural, unlike the derivative which is a linear transformation depends on no such choices.

10 Lecture-10

10.1 Chain Rule

The chain rule exists for differentiating a function of another function. If you want to learn more about the partial derivative of functions of functions, check out <https://math.stackexchange.com/questions/3266639/notation-for-partial-derivative-of-functions-of-functions/>.

Definition 10.1. The simplest form of the chain rule is for real-valued functions of one real variable. It states that if g is a function that is differentiable at a point c (i.e. the derivative $g'(c)$ exists) and f is a function that is differentiable at $g(c)$, then the composite function $f \circ g$ is differentiable at c , and the derivative is

$$(f \circ g)'(c) = f'(g(c)) \cdot g'(c).$$

The rule is sometimes abbreviated as

$$(f \circ g)' = (f' \circ g) \cdot g'.$$

If $y = f(u)$ and $u = g(x)$, then this abbreviated form is written in Leibniz notation as:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

The points where the derivatives are evaluated may also be stated explicitly:

$$\left. \frac{dy}{dx} \right|_{x=c} = \left. \frac{dy}{du} \right|_{u=g(c)} \cdot \left. \frac{du}{dx} \right|_{x=c}.$$

Example 10.2.

$$f(x) = (4x - 1)^{17}$$

$$\text{Let } u = 4x - 1.$$

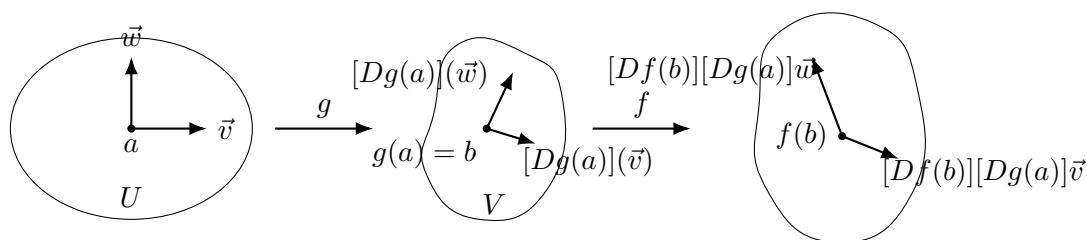
$$\frac{d}{dx}(u^{17}) = 17u^{16} \frac{du}{dx}, \quad \frac{du}{dx} = 4$$

$$f'(x) = 17(4x - 1)^{16} \cdot 4 = 68(4x - 1)^{16}$$

Remark: We could expand $(4x - 1)^{17}$ first and avoid the chain rule. Avoiding the chain rule is called the pain rule.

Theorem 10.3. Let $U \subset \mathbb{R}^n$, $V \subset \mathbb{R}^m$ be open sets, let $g : U \rightarrow V$ and $f : V \rightarrow \mathbb{R}^p$ be mappings, and let a be a point of U . If g is differentiable at a and f is differentiable at $g(a)$, then the composition $f \circ g$ is differentiable at a , and its derivative is given by

$$D(f \circ g)(a) = Df(g(a)) \circ Dg(a).$$



Proof. Will be added soon. □

There are three standard forms of the multivariable chain rule, distinguished by the dimensions of the domain and codomain.

1. $\mathbb{R} \xrightarrow{\varphi} \mathbb{R}^n \xrightarrow{f} \mathbb{R}$

Here

$$\varphi(t) = (x_1(t), \dots, x_n(t)), \quad f : \mathbb{R}^n \rightarrow \mathbb{R}.$$

Then $f \circ \varphi$ is a real-valued function of one variable:

$$(f \circ \varphi)(t) = f(x_1(t), \dots, x_n(t)).$$

By the chain rule,

$$\frac{d}{dt} f(\varphi(t)) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(\varphi(t)) x'_i(t).$$

This describes the derivative of a scalar field along a curve.

2. $\mathbb{R}^n \xrightarrow{g} \mathbb{R}^m \xrightarrow{f} \mathbb{R}$

Here

$$g = (g_1, \dots, g_m) : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad f : \mathbb{R}^m \rightarrow \mathbb{R}.$$

Then

$$(f \circ g)(x) = f(g_1(x), \dots, g_m(x)).$$

The chain rule gives

$$D(f \circ g)(a) = Df(g(a)) Dg(a).$$

In coordinates,

$$\frac{\partial (f \circ g)}{\partial x_j}(a) = \sum_{i=1}^m \frac{\partial f}{\partial y_i}(g(a)) \frac{\partial g_i}{\partial x_j}(a), \quad j = 1, \dots, n.$$

This is the scalar-valued multivariable chain rule.

3. $\mathbb{R}^n \xrightarrow{g} \mathbb{R}^m \xrightarrow{f} \mathbb{R}^l$

This is the most general case. Let

$$g : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad f : \mathbb{R}^m \rightarrow \mathbb{R}^l.$$

Then

$$f \circ g : \mathbb{R}^n \rightarrow \mathbb{R}^l.$$

The chain rule is

$$D(f \circ g)(a) = Df(g(a)) Dg(a).$$

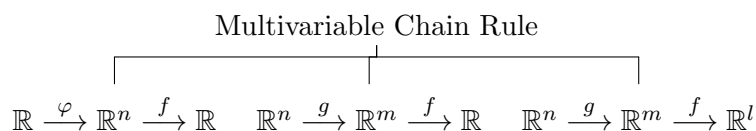
If

$$f = (f_1, \dots, f_l),$$

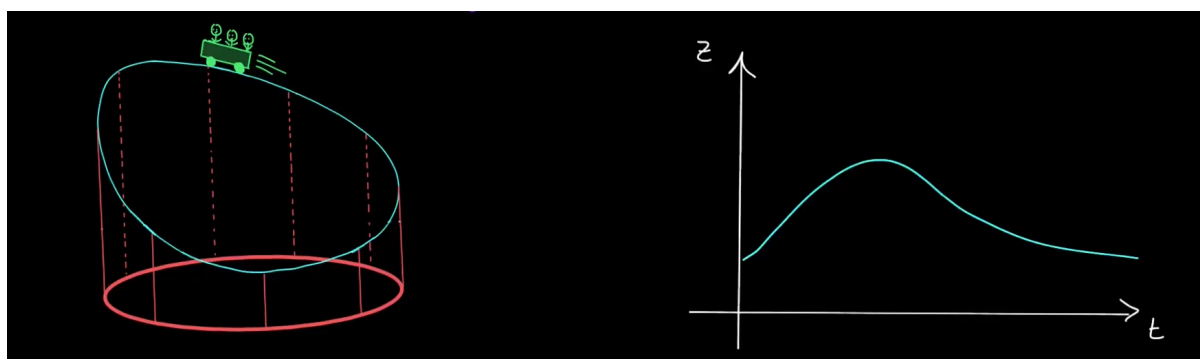
then componentwise

$$\frac{\partial(f_r \circ g)}{\partial x_j}(a) = \sum_{i=1}^m \frac{\partial f_r}{\partial y_i}(g(a)) \frac{\partial g_i}{\partial x_j}(a), \quad r = 1, \dots, l, \quad j = 1, \dots, n.$$

So the derivative of the composition is obtained by multiplying the Jacobian matrices.



Example 10.4. The (x, y) coordinates of a roller coaster at time t are given by $(\cos t, \sin t)$ and its height at (x, y) is given by $z = x^2y + x + 1$. What is its rate of change in height with respect to time at t ?



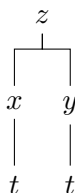
Approach-1: We can write the height as a function of time by substituting $x = \cos t$ and $y = \sin t$:

$$z(t) = (\cos t)^2 \sin t + \cos t + 1.$$

Then we can differentiate $z(t)$ with respect to t :

$$z'(t) = 2(\cos t)(-\sin t) \sin t + (\cos t)^2 \cos t - \sin t.$$

Approach-2: We can also use the multivariable chain rule.



Here, z is a function of x and y , and both x and y are functions of t . By the multivariable chain rule, we have

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$

Calculating the partial derivatives, we get

$$\frac{\partial z}{\partial x} = 2xy + 1, \quad \frac{\partial z}{\partial y} = x^2.$$

Calculating the derivatives of x and y with respect to t , we have

$$\frac{dx}{dt} = -\sin t, \quad \frac{dy}{dt} = \cos t.$$

Substituting these into the chain rule formula, we get

$$\frac{dz}{dt} = (2xy + 1)(-\sin t) + (x^2)(\cos t).$$

Finally, substituting $x = \cos t$ and $y = \sin t$, we find

$$\frac{dz}{dt} = (2(\cos t)(\sin t) + 1)(-\sin t) + (\cos^2 t)(\cos t).$$

Example 10.5. Given that,

$$F : \mathbb{R} \rightarrow \mathbb{R}^3$$

$$t \mapsto \begin{pmatrix} t^2 \\ t^4 \\ -t^3 \end{pmatrix}$$

and,

$$G : \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \mapsto \sin(3x_1 + 2x_2 + 5x_3)$$

Find $\frac{d}{dt}(G \circ F)|_{t=1}$.

Example 10.6. Given that,

$$F : \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

$$\begin{pmatrix} u \\ v \end{pmatrix} \mapsto \begin{pmatrix} u + v \\ u^2 \\ v^3 \end{pmatrix}$$

and,

$$G : \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto xy + z$$

Find all the partial derivatives of $G \circ F$ at $(u, v) = (1, 1)$.

Example 10.7. Given that,

$$F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$\begin{pmatrix} u \\ v \end{pmatrix} \mapsto \begin{pmatrix} u^2 + v \\ uv \end{pmatrix}$$

and,

$$G : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x + y \\ x - y \end{pmatrix}$$

Find $D(G \circ F)|_{\vec{x}=(1,1)^T}$. Check the partial derivatives $\frac{\partial}{\partial u}$ and $\frac{\partial}{\partial v}$ for the component functions are nothing but the entries of the $D(G \circ F)$.

Example 10.8. Suppose the temperature in space is given by

$$f \begin{pmatrix} x \\ y \\ z \end{pmatrix} = xyz^2 + e^{3xy-2z}$$

and the position of a bumblebee is given as a function of time by

$$g : \mathbb{R} \rightarrow \mathbb{R}^3.$$

If at time $t = 0$ the bumblebee is at

$$a = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

and her velocity vector is

$$\mathbf{v} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix},$$

then we might ask at what rate she perceives the temperature to be changing at that instant. The temperature she measures at time t is $f(g(t))$, and so she wants to calculate

$$(f \circ g)'(0) = Df(g(0))g'(0).$$

We have

$$Df \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{bmatrix} yz^2 + 3ye^{3xy-2z} & xz^2 + 3xe^{3xy-2z} & 2xyz - 2ze^{3xy-2z} \end{bmatrix},$$

so

$$Df(a) = \begin{bmatrix} 24 & 12 & 10 \end{bmatrix}.$$

Then

$$(f \circ g)'(0) = Df(g(0)) g'(0) = Df(a)\mathbf{v} = \begin{bmatrix} 24 & 12 & 10 \end{bmatrix} \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = 8.$$

Note that in order to apply the chain rule, we need to know only her position and velocity vector at that instant, not even what her path near a might be.

11 Lecture-11

11.1 Inverse Function Theorem

The phrase *inverse function* appears first in elementary mathematics in a set-theoretic form: a function

$$f : X \rightarrow Y$$

has an inverse if and only if it is bijective. This is correct, but in analysis and geometry it is not the whole story. For maps between Euclidean spaces,

$$f : \mathbb{R}^m \rightarrow \mathbb{R}^n,$$

we usually want more than merely an inverse as a rule of correspondence between sets. We want the inverse to preserve the analytic structure of the problem: continuity, differentiability, smoothness, and the local geometry of the spaces involved.

The inverse function theorem explains when a smooth map behaves *locally* like a reversible change of coordinates. Let $f : X \rightarrow Y$ be a function between sets.

Proposition 11.1. *The following are equivalent:*

- (1) f has an inverse function $f^{-1} : Y \rightarrow X$,
- (2) f is bijective,
- (3) for each $y \in Y$, there exists a unique $x \in X$ such that $f(x) = y$.

This statement is purely about matching elements. It does not know anything about: continuity, open sets, differentiability, tangent vectors, local coordinates. So from the point of view of analysis, the set-theoretic inverse is too weak. A bijection can be very irregular. For maps between spaces carrying more structure, there are different notions of invertibility:

- **Set-theoretic inverse:** f is bijective.
- **Topological inverse:** f is a homeomorphism.
- **Smooth inverse:** f is a diffeomorphism.

The inverse function theorem belongs to the smooth category. Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is smooth. Even if f is globally bijective, that still does not guarantee that f^{-1} is smooth.

Example 11.2. Consider

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^3.$$

This map is bijective and smooth. Its inverse is

$$f^{-1}(y) = \sqrt[3]{y}.$$

The inverse is continuous, but it is not differentiable at $y = 0$. Indeed,

$$\frac{d}{dy}(y^{1/3}) = \frac{1}{3y^{2/3}},$$

which is not defined at 0.

So even in the easiest case, global bijectivity is not enough for a smooth inverse. Something more must be required. The inverse function theorem is not a theorem about global invertibility. It is a theorem about *local* invertibility.

Example 11.3. Let

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^3 - x.$$

This function is not globally injective. However,

$$f'(x) = 3x^2 - 1.$$

At any point where $3x^2 - 1 \neq 0$, the function is locally invertible.

Thus local invertibility is controlled by the derivative, not by global behavior.

Let $f : U \rightarrow \mathbb{R}^n$ be differentiable, where $U \subseteq \mathbb{R}^n$ is open, and let $a \in U$. Then for small h ,

$$f(a + h) = f(a) + Df(a)h + \text{error},$$

where the error is small compared with $\|h\|$. So near a , the function behaves like its derivative $Df(a)$, which is a linear map

$$Df(a) : \mathbb{R}^n \rightarrow \mathbb{R}^n.$$

This leads to the fundamental idea:

To understand whether f is locally invertible near a , first ask whether the linear map $Df(a)$ is invertible.

If $Df(a)$ is invertible, then its action on small balls is easy to visualize: it may stretch, rotate, reflect, or shear, but it does not collapse any direction.

If $Df(a)$ is singular, then some infinitesimal direction is flattened, and local invertibility may fail.

Theorem 11.4 (Inverse Function Theorem). *Let $U \subseteq \mathbb{R}^n$ be open, and let*

$$f : U \rightarrow \mathbb{R}^n$$

be a C^1 map. Suppose that for some $a \in U$, the derivative $Df(a)$ is invertible. Equivalently,

$$\det Df(a) \neq 0.$$

Then there exist open neighborhoods $U_0 \subseteq U$ of a and $V_0 \subseteq \mathbb{R}^n$ of $f(a)$ such that

$$f : U_0 \rightarrow V_0$$

is a bijection, and its inverse

$$f^{-1} : V_0 \rightarrow U_0$$

is also C^1 .

Moreover,

$$D(f^{-1})(f(a)) = (Df(a))^{-1}.$$

The theorem says that invertibility of the derivative at one point forces invertibility of the nonlinear map on small neighborhoods.

Exercises

Example 11.5 (A linear map). Let

$$f(x, y) = (2x + y, x - y).$$

Then

$$Df(x, y) = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \quad \det Df(x, y) = -3 \neq 0.$$

Hence f is locally invertible everywhere. Since this is a linear map with an invertible matrix, it is in fact globally invertible.

Example 11.6 (Polar coordinates away from the origin). Consider

$$f(r, \theta) = (r \cos \theta, r \sin \theta), \quad r > 0.$$

Its Jacobian matrix is

$$Df(r, \theta) = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix},$$

so

$$\det Df(r, \theta) = r.$$

For $r > 0$, this is nonzero. Hence f is locally invertible wherever $r > 0$. However, f is not globally injective, because θ and $\theta + 2\pi$ determine the same point. Now, we can compute the derivative of the inverse function without even finding the inverse function explicitly,

$$\begin{aligned} D(f^{-1}) &= (Df)^{-1} \\ &= \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}^{-1} \\ &= \begin{pmatrix} \cos \theta & \sin \theta \\ -\frac{\sin \theta}{r} & \frac{\cos \theta}{r} \end{pmatrix} \\ &= \begin{pmatrix} \frac{x}{\sqrt{x^2 + y^2}} & \frac{y}{\sqrt{x^2 + y^2}} \\ -\frac{-y}{x^2 + y^2} & \frac{x}{x^2 + y^2} \end{pmatrix} \end{aligned}$$

This matches with the derivative of the inverse function, which can be computed by implicit differentiation.

$$f^{-1} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \sqrt{x^2 + y^2} \\ \tan^{-1} \frac{y}{x} \end{pmatrix}$$

Use this inverse function to compute the derivative $D(f^{-1})$.

Example 11.7 (A smooth bijection with non-smooth inverse). Let

$$f(x) = x^3.$$

Then f is bijective and smooth, but

$$f'(0) = 0.$$

The inverse is not differentiable at 0. So f is not a local diffeomorphism at the origin.

Example 11.8 (Failure when dimensions do not match). Define

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x, y) = x^2 + y^2.$$

Then

$$Df(x, y) = \begin{pmatrix} 2x & 2y \end{pmatrix},$$

which is a 1×2 matrix. There is no square Jacobian here, hence no usual local inverse between open subsets of Euclidean spaces.

Example 11.9. Consider

$$f(u, v) = (u^2 - v^2, 2uv).$$

Then

$$Df(u, v) = \begin{pmatrix} 2u & -2v \\ 2v & 2u \end{pmatrix}, \quad \det Df(u, v) = 4(u^2 + v^2).$$

So the map is locally invertible away from $(0, 0)$, but it fails precisely at the origin. This identifies the singular point immediately.

Example 11.10. Consider the map

$$f : \mathbb{R}^3 \setminus \{(x, y, z) : x+y+z+1 = 0\} \rightarrow \mathbb{R}^3, \quad f(x, y, z) = \left(\frac{x}{x+y+z+1}, \frac{y}{x+y+z+1}, \frac{z}{x+y+z+1} \right).$$

- Show that f satisfies the hypotheses of the inverse function theorem whenever $x+y+z+1 \neq 0$.
- Find f^{-1} .

Solution. Let

$$s = x + y + z + 1.$$

Then

$$f(x, y, z) = \left(\frac{x}{s}, \frac{y}{s}, \frac{z}{s} \right).$$

The Jacobian matrix of f is

$$Df(x, y, z) = \begin{pmatrix} \frac{y+z+1}{s^2} & -\frac{x}{s^2} & -\frac{x}{s^2} \\ -\frac{y}{s^2} & \frac{x+z+1}{s^2} & -\frac{y}{s^2} \\ -\frac{z}{s^2} & -\frac{z}{s^2} & \frac{x+y+1}{s^2} \end{pmatrix}.$$

Hence

$$\det Df(x, y, z) = \frac{1}{(x+y+z+1)^4} = \frac{1}{s^4}.$$

Therefore,

$$\det Df(x, y, z) \neq 0 \quad \text{whenever } x+y+z+1 \neq 0.$$

So f satisfies the conditions of the inverse function theorem at every point of its domain.

Now let

$$(X, Y, Z) = f(x, y, z).$$

Then

$$X = \frac{x}{x + y + z + 1}, \quad Y = \frac{y}{x + y + z + 1}, \quad Z = \frac{z}{x + y + z + 1}.$$

Write

$$f^{-1}(X, Y, Z) = (u, v, w).$$

Then

$$(X, Y, Z) = f(u, v, w) = \left(\frac{u}{u + v + w + 1}, \frac{v}{u + v + w + 1}, \frac{w}{u + v + w + 1} \right),$$

so

$$u = X(u + v + w + 1), \quad v = Y(u + v + w + 1), \quad w = Z(u + v + w + 1).$$

Adding these equations gives

$$u + v + w = (X + Y + Z)(u + v + w + 1).$$

Thus,

$$u + v + w = \frac{X + Y + Z}{1 - (X + Y + Z)},$$

and therefore

$$u + v + w + 1 = \frac{1}{1 - X - Y - Z}.$$

Substituting back, we obtain

$$u = \frac{X}{1 - X - Y - Z}, \quad v = \frac{Y}{1 - X - Y - Z}, \quad w = \frac{Z}{1 - X - Y - Z}.$$

Hence

$$f^{-1}(X, Y, Z) = \left(\frac{X}{1 - X - Y - Z}, \frac{Y}{1 - X - Y - Z}, \frac{Z}{1 - X - Y - Z} \right), \quad X + Y + Z \neq 1.$$

So the inverse map is

$$f^{-1}(x, y, z) = \left(\frac{x}{1 - x - y - z}, \frac{y}{1 - x - y - z}, \frac{z}{1 - x - y - z} \right), \quad x + y + z \neq 1.$$

12 Lecture-12

12.1 Implicit Function Theorem

Let's look $x^2 + y^2 = 1$ closely. Is this a graph of a function? It's not. No, it's not. It's a relation between x and y . But we want to study function so far. At a neighborhood of (x, y) except $(0, 1)$ and $(0, -1)$, we can solve for y as a function of x . $y = \sqrt{1 - x^2}$ or $y = -\sqrt{1 - x^2}$ are explicit functions that give the same relation between x and y as $x^2 + y^2 = 1$. This is true locally near any point except $(0, 1)$ and $(0, -1)$.

The implicit function theorem provides conditions under which a relationship of the form $F(x, y) = 0$ can be rewritten as a function $y = f(x)$ locally.

Example 12.1. Consider $y^5 + y^3 + y + x = 0$. It is trivial to write $x = f(y)$, but can we write $y = f(x)$? But there is no explicit formula for y in terms of x , because there is no formula for the roots of a general quintic polynomial. In general, there is no formula for finding roots of polynomials of degree 5 or higher (Abel, Galois). Fix x_0 , define $\varphi(y) = y^5 + y^3 + y + x_0$. Then,

$$\varphi'(y) = 5y^4 + 3y^2 + 1 > 0, \quad \forall y \in \mathbb{R}.$$

Which implies φ is strictly increasing. In Calculus-I, we know that any odd degree polynomial has at least one real root. But since φ is strictly increasing, it has exactly one real root. So we can write $y = f(x)$ locally near any point (x_0, y_0) , give me x_0 , we can find a unique y_0 such that $y_0^5 + y_0^3 + y_0 + x_0 = 0$. We found a function $y = f(x)$, but f is implicit.

Theorem 12.2. Denote $\vec{x} = (x_1, \dots, x_n)$. Let $F(\vec{x}, y) \in C^1$ in a neighborhood of $N_0(\vec{x}_0, y_0)$ such that

1. $F(x_0, y_0) = 0$,
2. $\frac{\partial F}{\partial y}(x_0, y_0) \neq 0$.

Then there exists a neighborhood of (x_0, y_0) in which there is a unique (implicit) function $y = f(\vec{x})$ such that

- (i) $y_0 = f(\vec{x}_0)$,
- (ii) $F(\vec{x}, f(\vec{x})) = 0$ for every x in the neighborhood,
- (iii) And the derivative of f is given by,

$$\frac{\partial f}{\partial x_i} = -\frac{\frac{\partial F}{\partial x_i}}{\frac{\partial F}{\partial y}}$$

in the neighborhood of $(\vec{x}_0, y_0)$.

Example 12.3. Given that $3x^2y - yz^2 - 4xz - 7 = 0$. We will show that near $(-1, 1, 2)$ we can write $y = f(x, z)$, and we will find $\frac{\partial f}{\partial x}(-1, 2)$. Here,

$$F(x, y, z) = 3x^2y - yz^2 - 4xz - 7 \in C^1$$

And $F(-1, 1, 2) = 0$. Moreover,

$$\frac{\partial F}{\partial y} = 3x^2 - z^2, \quad \frac{\partial F}{\partial y}(-1, 1, 2) = -1 \neq 0.$$

Thus, by the implicit function theorem, there exists a neighborhood of $(-1, 1, 2)$ in which there is a unique function $y = f(x, z) \in C^1$ such that $f(-1, 2) = 1$ and $F(x, f(x, z), z) = 0$ for every (x, z) in the neighborhood. Moreover,

$$\frac{\partial f}{\partial x}(-1, 2) = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}}(-1, 1, 2) = -\frac{6xy - 4z}{3x^2 - z^2}(-1, 1, 2) = -\frac{6(-1)(1) - 4(2)}{3(-1)^2 - (2)^2} = -\frac{-6 - 8}{3 - 4} = -\frac{-14}{-1} = -14.$$

In fact,

$$\begin{aligned} 3x^2y - yz^2 &= 4xz + 7 \\ y(3x^2 - z^2) &= 4xz + 7 \\ y &= \frac{4xz + 7}{3x^2 - z^2}. \end{aligned}$$

In this example, we can explicitly solve for y in terms of x and z , but the implicit function theorem guarantees the existence of such a function even when an explicit formula is not available. The key point is that the condition $\frac{\partial F}{\partial y}(-1, 1, 2) \neq 0$ ensures that we can locally solve for y as a function of x and z . In this example, we can write $z = f(x, y)$ as well, but the theorem fails at $(-1, 1, 2)$ because

$$\frac{\partial F}{\partial z}(-1, 1, 2) = 0.$$

Example 12.4.

$$\begin{aligned} \text{ex.} \quad F(x, y) &= (x - y)^3 \\ F(x, y) = 0 &\iff y = x \end{aligned}$$

There is an explicit function at any point. However, at $(0, 0)$,

$$\frac{\partial F}{\partial x}(0, 0) = 3(x - y)^2|_{(0,0)} = 0,$$

$$\frac{\partial F}{\partial y}(0, 0) = -3(x - y)^2|_{(0,0)} = 0.$$

The theorem doesn't apply.

Let's prove the implicit function theorem for the case of two variables only.

Proof. $\frac{\partial F}{\partial y}(x_0, y_0) \neq 0$ implies w.l.o.g. $\frac{\partial F}{\partial y}(x_0, y_0) > 0$. Since, $F \in C^1$, $\frac{\partial F}{\partial y}$ is continuous, so $\frac{\partial F}{\partial y} > 0$ at a neighborhood of (x_0, y_0) . What do we know about $F(x_0, y)$ function? Here we fixing x_0 , and view F as a function of y . We know that $\frac{\partial F}{\partial y} > 0$ at a neighborhood of (x_0, y_0) , which implies $F(x_0, y)$ is a strictly increasing function. $\exists y_1$ such that $F(x_0, y_1) > 0$ and $\exists y_2$ such that $F(x_0, y_2) < 0$. Since $F \in C^1$, F is continuous. Which implies $\forall x$ near x_0 (releasing our first variable),

$$F(x, y_1) > 0, \quad F(x, y_2) < 0$$

For such an x near x_0 , since $\frac{\partial F}{\partial y} > 0$, (x, y) is strictly increasing as a function of y , therefore $\exists y$

such that $F(x, y) = 0$. This proves that $y = f(x)$ exists and is unique. Now,

$$F(x, f(x)) = 0$$

$$\frac{\partial F}{\partial x} \cdot 1 + \frac{\partial F}{\partial y} \frac{df}{dx} = 0 \implies \frac{df}{dx} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}}.$$

□

Theorem 12.5.

$$\begin{cases} F_1(x_1, \dots, x_n, z_1, \dots, z_m) &= 0 \\ \vdots \\ F_m(x_1, \dots, x_n, z_1, \dots, z_m) &= 0 \end{cases}$$

where $F_i \in C^1$ for $i = 1, \dots, m$. Let

$$\Delta = \det \begin{pmatrix} \frac{\partial F_1}{\partial z_1} & \dots & \frac{\partial F_1}{\partial z_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial z_1} & \dots & \frac{\partial F_m}{\partial z_m} \end{pmatrix}$$

If $\Delta \neq 0$ at a point $(\bar{x}^0, \bar{z}^0)$, then near that point, there are unique functions $z_i = f_i(x_1, \dots, x_n)$ for $i = 1, \dots, m$ such that the system of equations is satisfied. and the partial derivatives of f_i 's can be computed using implicit differentiation.

Example 12.6. Consider the system of equations:

$$\begin{cases} xu + yvu^2 &= 2 \\ xu^3 + y^2v^4 &= 2 \end{cases}$$

Let's show that near $(x, y, u, v) = (1, 1, 1, 1)$, we can express u and v as functions of x and y .

$$\begin{cases} F_1(x, y, u, v) &= xu + yvu^2 - 2 \\ F_2(x, y, u, v) &= xu^3 + y^2v^4 - 2 \end{cases}$$

Here,

$$\Delta = \det \begin{pmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} \end{pmatrix} = \det \begin{pmatrix} x + 2yvu & yu^2 \\ 3xu^2 & 4y^2v^3 \end{pmatrix}.$$

Evaluating at the point $(1, 1, 1, 1)$ gives:

$$\Delta(1, 1, 1, 1) = \det \begin{pmatrix} 1 + 2(1)(1) & 1(1)^2 \\ 3(1)(1)^2 & 4(1)^2(1)^3 \end{pmatrix} = \det \begin{pmatrix} 3 & 1 \\ 3 & 4 \end{pmatrix} = 3 \cdot 4 - 1 \cdot 3 = 12 - 3 = 9 \neq 0.$$

By the implicit function theorem, there exist unique functions $u = f_1(x, y)$ and $v = f_2(x, y)$ near the point $(1, 1, 1, 1)$. Let's find $\frac{\partial u}{\partial x}(1, 1)$:

$$\begin{cases} u + x \frac{\partial u}{\partial x} + y \frac{\partial v}{\partial x} u^2 + 2yvu \frac{\partial u}{\partial x} &= 0 \\ u^3 + 3xu^2 \frac{\partial u}{\partial x} + y^2 \cdot 4v^3 \frac{\partial v}{\partial x} &= 0 \end{cases} \implies \begin{cases} 3 \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} &= -1 \\ 3 \frac{\partial u}{\partial x} + 4 \frac{\partial v}{\partial x} &= -1 \end{cases}$$

Now, solve the system of equations to find $\frac{\partial u}{\partial x}(1, 1)$.

References

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